

Understanding continuance intention toward the use of AI chatbots in customer service among generation Z in Vietnam

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ABSTRACT

As artificial intelligence (AI) becomes integral to digital customer service, AI chatbots increasingly shape user experience and operational efficiency. This research examines the factors influencing Generation Z's continuance intention to use AI chatbot customer services in Vietnam by integrating the Information System Success Model (ISSM) with chatbot marketing effort constructs, including interaction, customization, problem-solving, and perceived enjoyment. Data from 562 valid responses were collected using a non-probability convenience sampling method and analyzed with Partial Least Squares Structural Equation Modeling (PLS-SEM). The results show that system quality, information quality, service quality, and perceived enjoyment significantly influence both user attitude and satisfaction, with system quality having the strongest effect. Customization and interaction positively affect only attitude, while problem-solving positively influences only satisfaction. Attitude and satisfaction, in turn, strongly predict continuance intention. These findings suggest that while experiential features primarily influence user attitude, satisfaction is driven by functional performance. This underscores the importance of integrating technical reliability with emotionally engaging, user-centered design to support sustained AI chatbot use. Theoretically, the study advances the ISSM by incorporating affective and behavioral dimensions, offering a richer understanding of post-adoption behavior among digital-native users. Practically, it provides actionable guidance for developers and marketers to enhance chatbot effectiveness through personalization, intuitive usability, and emotionally appealing experiences tailored to Generation Z.

1. Introduction

Digital transformation is increasingly regarded as a pivotal trend in customer service, where technologies such as chatbots serve a crucial function in improving user experiences and streamlining business processes. However, challenges remain, including the pace of transformation, cultural contexts, and the roles of middle management (Nadkarni & Prügl, 2020). As consumer expectations for faster, seamless, and personalized service continue to intensify, businesses are compelled to adopt advanced technologies. Among them, chatbots have emerged as a central solution to meet these demands by enabling efficient, responsive, and individualized customer interactions (Kaushal & Yadav, 2022).

The concept of chatbots dates to 1966 with the creation of ELIZA by Joseph Weizenbaum, which laid the foundation for computer-based

conversation systems. Since then, chatbot technology has evolved dramatically. Fueled by breakthroughs in AI, modern chatbots have reached new levels of sophistication (Wiederhold, 2023). According to Botpress (2024), the global chatbot market is projected to reach \$27.3 billion by 2030, reflecting its growing importance across industries. This trend underscores the role of chatbots in automating interactions, improving user experiences, and delivering scalable, cost-effective, and intelligent solutions for customer engagement (Botpress, 2024).

The increasing deployment of AI chatbots is driven by the convergence of natural language processing (NLP), real-time speech recognition, and sentiment analysis, enabling intelligent, emotionally responsive, and multilingual customer interactions (Dash et al., 2023; Jha et al., 2024). Despite this growth, there are still concerns about security risks and weaknesses (Hasal et al., 2021). AI chatbots today can simulate natural human conversations, which makes them increasingly

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prevalent in industries such as e-commerce, travel, tourism, and healthcare, where interactive and high-quality engagement is essential (Kurniawan et al., 2024).

Advancements in AI technology have elevated chatbots into powerful tools that surpass traditional customer service models by enabling personalized, automated interactions. These AI-driven systems offer numerous advantages, including 24/7 availability, tailored user experiences, scalability, cost efficiency, data-driven insights, and unbiased engagement (Aggarwal et al., 2023; Meshram et al., 2021). By operating continuously without the need for human oversight, AI chatbots ensure seamless customer interaction and deliver highly customized responses through behavioral data analysis. In contrast to conventional chatbots, which are constrained by static scripts and scalability challenges, AI chatbots can handle large volumes of inquiries efficiently while substantially reducing operational costs (Adiguzel et al., 2023; Zhang et al., 2020). Furthermore, their capacity to analyze large datasets using machine learning (ML) enables real-time insights that enhance decision-making and build user trust. This advantage is particularly valuable in emotionally sensitive domains, such as mental health and behavioral support, where traditional systems are often constrained by human biases or limitations (Androutopoulou et al., 2018).

In recent years, AI chatbots have garnered increased scholarly interest due to their ability to automate routine tasks, enhance user engagement, and transform service delivery across diverse sectors such as education, healthcare, and customer support, thereby improving efficiency and accessibility (Følstad et al., 2021). Increasing research has explored how factors such as information quality, service quality, perceived enjoyment, usefulness, and ease of use shape user satisfaction and long-term engagement (Ashfaq et al., 2020). For instance, Hui et al. (2023) demonstrated that the quality of information and service quality positively affect user satisfaction, which in turn strengthens continuance intention toward AI chatbot use. This aligns with findings from Hsu and Lin (2022), who revealed that AI chatbots' service quality has a strong influence on user satisfaction in digital service contexts. Moreover, Chen et al. (2023) affirmed that the service quality of AI chatbots significantly contributes to fostering customer loyalty to an organization through satisfaction.

Regional studies also support these conclusions by examining the application of chatbots in specific industries. Zakaria et al. (2024) explored chatbot use in Palestine's e-commerce sector and found that service quality positively influenced user satisfaction. In the broader context of chatbot marketing, studies have emphasized the role of AI-powered chatbots as critical tools for strengthening customer-brand relationships and shaping customer reactions (Cheng & Jiang, 2021). Similarly, Xia and Shannon (2024) identified that key elements of chatbot marketing, specifically interaction quality, perceived enjoyment, customization, and problem-solving capabilities, positively and significantly influence perceived ease of use, perceived usefulness, user attitudes, and the customer-brand relationship in the context of AI-powered chatbots. In education, Pasupuleti and Thiyyagura (2024) confirmed that students' satisfaction with ChatGPT is significantly influenced by system, information, service, and conversational quality, which subsequently drives their continuance intention. Additionally, in online retail, AI chatbots improve responsiveness and usability, which in turn increase both extrinsic and intrinsic value perceptions, foster greater customer satisfaction, and strengthen brand affinity (Chen, Le, & Florence, 2021). During the COVID-19 pandemic, Zhu et al. (2022) examined the use of chatbots for mental health services in China, underscoring the importance of personalization and enjoyment. However, limitations in effectiveness also emerged, highlighting the need for better contextual tailoring.

In Vietnam, the adoption of AI chatbots is expanding rapidly and drawing academic attention. Nguyen and Le (2024) employed the Information Systems Success Model (ISSM) to investigate how the service attributes of chatbots impact customer experience in the digital banking sector. Their findings indicated that chatbot attributes shape extrinsic

and intrinsic values, which significantly predict user satisfaction and continuance intention in banking services. Likewise, Duong et al. (2024) studied ChatGPT among Vietnamese students and found that information and service quality positively impacted trust, satisfaction, and continuance intention. However, these studies have largely focused on technical functionality and core ISSM constructs, offering limited insight into users' affective experiences and marketing-driven engagement factors.

Despite increasing interest in AI chatbot adoption, notable theoretical and contextual gaps persist. In the Vietnamese context, most existing studies concentrated on traditional system-oriented attributes, such as system quality, information quality, and service quality, while giving limited attention to affective and marketing-driven features. Key experiential constructs, including interaction quality, perceived enjoyment, customization, and problem-solving capabilities, which play a crucial role in sustaining user engagement, remain underexplored. Elements such as emotional satisfaction, personalized responses, and interactive communication have become central to effective human-AI interaction, yet these dimensions are not adequately addressed within the conventional ISSM frameworks commonly applied in Vietnam. These gaps are especially relevant when considering the behavioral patterns of Generation Z, a digitally fluent and highly connected demographic that grew up alongside internet technologies, social media, and AI-driven systems. As lifelong users of digital platforms, Generation Z not only dominates current digital consumption but also holds the key to shaping future market trends, making them a vital group for examining AI chatbot adoption in customer service. According to PwC Vietnam's 2020 survey on digital readiness, 84 % of Generation Z reported positive attitudes toward technology in their professional lives (PricewaterhouseCoopers, 2020). Moreover, by 2025, they are expected to account for nearly one-third of Vietnam's total workforce, further underscoring their socioeconomic significance (PricewaterhouseCoopers, 2020). With distinct digital habits, strong preferences for personalization, and a reliance on interactive technologies, Generation Z increasingly demands more adaptive, intuitive, and emotionally engaging digital experiences. Accordingly, a focused investigation into this demographic is both timely and necessary to identify the key drivers of their satisfaction and continuance intention regarding AI chatbot use. Gaining a deeper understanding of their expectations and behavioral patterns is essential for guiding the design and delivery of AI chatbot services in Vietnam's rapidly evolving digital ecosystem.

To address existing research gaps, this study investigates the key factors that drive the ongoing use of AI chatbots among Vietnam's Generation Z, a digitally native demographic that remains underexplored in the context of chatbot adoption. Recognizing the evolving expectations of this tech-savvy cohort, this study aims to examine both technical and experiential determinants of user attitude, satisfaction, and continuance intention. Grounded in the ISSM, this research extends the theoretical framework by incorporating key marketing-oriented constructs, namely interaction, customization, problem-solving, and perceived enjoyment. This extension enables a more holistic examination of both the technical performance and emotional experience of chatbot use, thereby advancing understanding of post-adoption behavior in AI-driven customer service.

The novelty of this study is reflected in its integrated contributions across theory, context, and practice. Firstly, it extends the ISSM by incorporating affective and behavioral constructs that capture the emotional and experiential aspects of human-AI interaction, dimensions that have been largely overlooked in prior applications of the model within the Vietnamese context. Secondly, the study offers contextual relevance by focusing on Generation Z in Vietnam, a digitally native cohort whose distinct preferences, behaviors, and expectations differ significantly from previous generations, thereby warranting targeted investigation. Thirdly, the research generates practical insights for developers and marketers aiming to improve AI chatbot adoption by emphasizing the need to align technical functionality with personalized,

engaging, and socially intelligent user experiences. By addressing both cognitive and affective factors influencing chatbot use, this study provides a more comprehensive understanding of post-adoption behavior. Its findings contribute to the theoretical advancement of human-computer interaction and deliver actionable recommendations for enhancing chatbot design, fostering user satisfaction, and sustaining engagement in digital service environments.

2. Literature review

2.1. Theoretical background

2.1.1. Artificial intelligence (AI) chatbots

In the current digital era, Artificial Intelligence (AI) has emerged as a transformative catalyst, reshaping numerous fields, revolutionizing traditional practices, and enabling new capabilities (Brem et al., 2021; Chen, Chiu, et al., 2021). Among the various applications of AI, chatbots stand out as some of the most advanced implementations, gaining attention for their ability to simulate human-like conversations and provide quick, simple responses across domains such as commerce, education, and tourism (Benaddi et al., 2024). AI chatbots are software systems designed to mimic human dialogue through text or voice interfaces (Rane et al., 2022). These systems leverage advancements in Natural Language Processing (NLP), Machine Learning (ML), and related AI subfields to enable human-like interaction capabilities across various sectors (Benaddi et al., 2024). Recent developments in deep learning and the incorporation of bionic algorithms have significantly enhanced the contextual understanding and response generation capabilities of AI chatbots, allowing them to produce more coherent, natural, and human-like interactions (Pradhan et al., 2025).

The development of AI chatbot can be traced back to the creation of ELIZA in the 1960s by Joseph Weizenbaum at the Massachusetts Institute of Technology (MIT), a pioneering effort that laid the foundation for subsequent advancements in conversational AI (Singh & Thakur, 2020). Since then, chatbot technology has progressed rapidly, becoming increasingly intelligent, adaptive, and integrative across various digital platforms.

In contemporary practice, AI chatbots have become indispensable tools in fields such as marketing, e-commerce, and customer service (Gupta et al., 2024; Shumanov & Johnson, 2020). In marketing, AI chatbots are utilized to deliver personalized customer experiences, strengthen engagement, and foster deeper brand-consumer relationships (Shumanov & Johnson, 2020). In e-commerce, AI chatbots facilitate seamless customer journeys by providing efficient, real-time, and personalized support, ensuring continuous availability and effective resolution of customer inquiries (Gupta et al., 2024). In customer service, AI chatbots improve operational performance by reducing response times, offering individualized attention, and enhancing user satisfaction (Ashfaq et al., 2020). Cheng and Jiang (2021) highlighted the strategic significance of AI-driven chatbot marketing efforts in fostering and strengthening customer-brand relationships, underscoring their potential to support long-term engagement. Gupta et al. (2024) further emphasized the pivotal role of AI in revolutionizing e-commerce, particularly through enhanced personalization.

2.1.2. Information system success model (ISSM)

The Information Systems Success Model (ISSM), initially introduced by DeLone and McLean (1992) and later refined in 2003, provides a comprehensive framework for evaluating the effectiveness of information systems across diverse technological contexts (DeLone & McLean, 1992, 2003). The model has been extensively utilized in domains such as e-commerce, electronic services, travel assistance, digital banking, and AI chatbots (Ashfaq et al., 2020; Dirgantari et al., 2020; Duong et al., 2024; Foroughi et al., 2024; Nguyen & Le, 2024; Orden-Mejía et al., 2024). Its adaptability and structured approach make it a valuable tool for systematically analyzing system success and user outcomes. In the

domain of AI-powered conversational agents, the ISSM has gained increasing relevance as a theoretical lens to evaluate user experience and system effectiveness. Researchers have employed the model to examine key success factors such as information quality, perceived enjoyment, and perceived usefulness in AI-driven customer interactions (Orden-Mejía et al., 2024). Overall, the model enables a sophisticated comprehension of how chatbot design and performance contribute to user engagement, trust, and long-term usage.

The ISSM model's constructs, including system quality, information quality, and service quality have been shown to significantly affect user satisfaction and continuance intention in digital environments. In the e-commerce sector, Foroughi et al. (2024) empirically confirmed that information quality, service quality, and system quality significantly influence users' perceptions, which subsequently enhance their attitudes, satisfaction, and continuance intention toward chatbot usage. Similarly, in e-service, ISSM has been employed by Ashfaq et al. (2020) to assess user satisfaction and continuance intention with AI-powered service agents. Their findings highlight the importance of both information quality and service quality in shaping user satisfaction, which subsequently serve as a key determinant of users' intention to continue engaging with chatbot services. These results further demonstrate the model's relevance for evaluating AI chatbot success across both functional and emotional dimensions of user interaction. Furthermore, the ISSM has proven valuable in evaluating AI chatbots' deployment in customer service, such as providing tourists with information about the city, where information quality, perceived enjoyment, and perceived usefulness positively affect satisfaction, which in turn strongly predicts continued chatbot use and revisit intention (Orden-Mejía et al., 2024). In Vietnam's digital banking, Nguyen and Le (2024) found that the ISSM is particularly well-suited for analyzing chatbot usage, as it clarifies the interrelationships among information quality constructs, satisfaction, and continuance intention, thereby providing comprehensive insights into user engagement in AI-enabled financial services. The model's widespread applicability and empirical validation across multiple sectors affirm its effectiveness in guiding the development, assessment, and optimization of AI chatbot systems. However, these studies have predominantly focused on system performance and functional quality, offering limited insights into the experiential or emotional aspects of human-AI interaction.

This study advances the ISSM by explicitly integrating affective and behavioral constructs, including interaction, customization, problem-solving, and perceived enjoyment, into the model. In contrast to prior research conducted in Vietnam, which focused narrowly on functional dimensions, our extended framework captures how experiential factors shape both attitude and satisfaction, providing a dual-pathway explanation of continuance intention. The inclusion of these chatbot-specific marketing effort variables allows for a more nuanced understanding of post-adoption behavior, particularly relevant for the Generation Z digital-native cohort, who value personalization, emotional engagement, and seamless digital interaction. This study contributes a novel, context-specific extension of ISSM by bridging technical and experiential dimensions, offering a more comprehensive explanation of sustained chatbot usage in emerging digital service environments.

2.2. Hypotheses development and research framework

2.2.1. Impact of system quality on attitude and satisfaction

System quality represents the technical soundness and functional efficiency of an information system, typically characterized by dimensions such as ease of use, functional capability, dependability, adaptability, data integrity, portability, system integration, and perceived importance (DeLone & McLean, 2003). Seddon (1997) noted that system quality includes the maintainability of program code, suggesting that clear and well-documented code structures are important for long-term system maintenance and enhancement. In their earlier work, DeLone and McLean (1992) identified reliability, flexibility,

responsiveness, and accuracy as fundamental components of system quality. More recently, [Foroughi et al. \(2024\)](#) argued that these attributes help reduce user frustration by minimizing conversational interruptions and lowering the cognitive load required to adapt to system-specific characteristics.

System quality exerts a positive influence on user attitude by ensuring that interactions with AI chatbots proceed without technical impediments. [Roh et al. \(2022\)](#) identified system quality as a key determinant of trust in fintech systems, which subsequently shaped users' attitudes toward those systems. This is consistent with findings in the chatbots context, where reliable and uninterrupted interactions support user confidence and favorable affective responses. Similarly, [Ahmed et al. \(2024\)](#) demonstrated that system quality, together with privacy considerations, exerts a significant positive effect on user attitudes, which in turn foster their behavioral intention to adopt voice-based AI assistant services in emerging economies. These findings align with broader information systems research, emphasizing that system quality not only shapes immediate attitudes but also sustains engagement by reinforcing trust and reliability in user-centric designs. In a recent study, [Foroughi et al. \(2024\)](#) showed that flexibility in handling varied user queries by chatbots reassures users of the system's adaptability to individual needs. Moreover, [Rahi et al. \(2025\)](#) demonstrated that system quality, along with other key constructs, significantly explain user attitude toward the adoption of AI-enabled service robots. Through these mechanisms, system quality shapes both affective and cognitive components of attitude and thus encourages sustained engagement with AI chatbots.

In addition, system quality emerges as a crucial driver of user satisfaction. According to [Kim and Kim \(2025\)](#), system quality positively influences the quality of AI-generated recommendations, thereby contributing to enhanced user satisfaction. When error rates are low and requests are handled promptly, users are less likely to experience frustration, leading to a smoother and more gratifying interaction experience. [Elayat and Elalfy \(2025\)](#) found that system-related aspects of AI chatbots' quality, such as system compliance, significantly enhance young consumers' satisfaction. In the banking sector, [Mulyono and Sfenrianto \(2022\)](#) discovered that system quality has a significant positive impact on customer satisfaction with chatbot services, indicating that system quality plays a crucial role in shaping users' overall satisfaction. These insights underscore that the ability of AI chatbots to deliver seamless, efficient, and contextually relevant interactions reduces cognitive effort and enhances perceptions of system competence. By fostering reliability and emotional assurance, system quality thus emerges as a pivotal driver of user satisfaction and sustained engagement with AI chatbots.

Accordingly, the hypotheses are formulated as follows:

- H1.** System Quality has a positive impact on Attitude toward AI chatbots.
- H2.** System Quality has a positive impact on Satisfaction with AI chatbots.

2.2.2. Impacts of information quality on attitude and satisfaction

Information quality is defined as the degree to which the outputs produced by a system are accurate, relevant to the user's context, timely, and capable of supporting effective decision-making ([Nelson et al., 2005](#); [Wixom & Todd, 2005](#)). In digital environments, high-quality information is fundamental to fostering user trust and sustained engagement. [Nicolaou et al. \(2012\)](#) argued that users' assessments of information exchange, particularly its accuracy and contextual alignment, contribute to both competence trust and goodwill trust, which subsequently influence perceived exchange and relationship risks. [Aziz et al. \(2024\)](#) described information quality as the extent to which system outputs are relevant, understandable, accurate, concise, complete, current, quickly available, and timely, all of which contribute to supporting users' information needs and usability. These attributes form the

foundation of users' perceptions of chatbots' effectiveness. [Mofokeng \(2021\)](#) further emphasized the importance of presenting quality information on websites, highlighting its significant role in building customer loyalty.

Information quality is widely recognized as a key antecedent of users' attitudes toward AI chatbots. Empirical evidence by [Kim, Li, and Choi \(2023\)](#) confirms that when chatbots deliver accurate, relevant, and timely information, users tend to form more favorable attitudes toward the service. Additionally, [Kwangsawad and Jattamart \(2022\)](#) discovered a strong relationship between perceived information quality and users' attitudes toward the continued use of community-enterprise customer-service chatbots. Their finding reinforces the idea that multidimensional information quality is essential for fostering positive attitudes toward AI chatbots. Furthermore, [Yadav et al. \(2023\)](#) emphasized the significance of information quality in fostering positive attitudes toward the sustained use of chatbots, asserting that providing high-quality information is fundamental for both initial adoption and ongoing engagement with AI chatbots. Collectively, these attributes of information quality enhance both the emotional and rational dimensions of user attitude. By addressing the informational needs of users through accuracy, completeness, and consistency, AI chatbots can significantly improve user experience and encourage continued interaction.

Information quality is an important factor determining user satisfaction with AI chatbots, as it ensures that the information provided aligns with users' expectations of accuracy, timeliness, and relevance. [Kim and Kim \(2025\)](#) found that high information quality positively influenced the quality of AI-generated recommendations, which, in turn, enhanced user satisfaction. This implies that when users receive accurate, relevant, and timely information, they are more likely to perceive the system as useful and reliable, thereby increasing their overall satisfaction. Their study demonstrated that enhancements in these areas lead to higher user satisfaction, while deficiencies often result in user workarounds, indicating dissatisfaction and system inefficiency. These findings underscore the centrality of information quality in determining how users perceive, evaluate, and engage with AI-enabled information systems. Moreover, [Niu and Mvondo \(2023\)](#) found that information quality significantly increases satisfaction in AI chatbot adoption, with accuracy and clarity emerging as key drivers of positive user experiences. [Kalankesh et al. \(2020\)](#) further confirmed that information quality is one of the core elements influencing user satisfaction, emphasizing that its careful integration in system development can enhance evaluations of information systems by enabling informed decision-making and fostering a sense of empowerment. Moreover, [Juquelier et al. \(2024\)](#) reported that perceptions of information quality are positively associated with customer satisfaction, although contextual factors such as time pressure may moderate this effect. This relationship can be explained by the fact that comprehensive, user-centric, and clearly delivered information fosters a sense of empowerment and trust, which in turn enhances user satisfaction and promotes long-term use of AI chatbots.

Accordingly, the hypotheses are formulated as follows:

- H3.** Information Quality has a positive impact on Attitude toward AI chatbots.
- H4.** Information Quality has a positive impact on Satisfaction with AI chatbots.

2.2.3. Impacts of service quality on attitude and satisfaction

Service quality in information systems refers to the overall support provided by the system or its personnel, emphasizing effectiveness, efficiency, and responsiveness in helping users achieve their goals ([DeLone & McLean, 2003](#)). It has been recognized as a key component in evaluating user experience, directly influencing users' perceptions of system usefulness and overall satisfaction. In the domain of customer service chatbots, [Ranieri et al. \(2024\)](#) observed that for task-oriented users, a deterioration in perceived service quality intensifies shopping-

related stress, which in turn increases the likelihood of journey abandonment.

High service quality enhances users' attitudes toward AI chatbots by providing responsive, empathetic, and reliable support that increases trust and perceived value. Chowdhury (2023) found that service quality significantly influences consumers' attitudes and behavioral intentions within the scope of digital food ordering services, emphasizing its central role in shaping user perceptions and decision-making processes. Similarly, Rupali et al. (2024) demonstrated that service quality exerts a direct and significant impact on users' attitudes toward AI chatbots. Their findings further reveal that customer loyalty is largely driven by user satisfaction and attitude, thereby underscoring the mediating roles these constructs play in promoting sustained user engagement with chatbot services. Moreover, the study by Maalouf et al. (2024) indicated that superior service quality contributed to the development of favorable user attitudes toward digital services. These findings underscore that high service quality contributes to positive user attitudes by reinforcing both trust and perceived usefulness in the chatbot's performance.

Service quality is a key contributor to user satisfaction with AI chatbots by ensuring reliable, efficient, and user-oriented service experiences. Chen et al. (2022) empirically validated that several key dimensions of AI chatbots' service quality positively influence user satisfaction, affirming the value of consistent and trustworthy chatbots' interactions. Besides that, Chang et al. (2025) revealed that service quality is a significant determinant of user satisfaction in digital banking. Their findings highlight that responsiveness and personalized attention, as key dimensions of service quality, play a crucial role in shaping users' emotional evaluations and, in turn, reinforcing the credibility of chatbots. Mehroli et al. (2023) found that perceived service quality dimensions significantly influence users' satisfaction and continued usage of banking chatbots, emphasizing the critical role of reliable and interactive journeys in fostering sustained user involvement. Furthermore, empirical evidence supports the notion that service quality enhances satisfaction by enabling smooth, efficient, and engaging user experiences (Chen et al., 2022). Overall, a high level of service quality not only meets functional needs but also promotes emotional fulfillment, thereby enhancing overall satisfaction in the AI chatbots' interaction.

Accordingly, the hypotheses are formulated as follows:

H5. Service Quality has a positive impact on Attitude toward AI chatbots.

H6. Service Quality has a positive impact on Satisfaction with AI chatbots.

2.2.4. Impact of interaction on attitude and satisfaction

In the domain of conversational interaction, interaction is understood as the dynamic, bidirectional communication between a user and a chatbot, whereby users can send messages and receive information or content (Haugeland et al., 2022; Li et al., 2020). This concept extends beyond simple message transmission, encompassing dialogue structures, communication modes, and interaction styles (social-oriented vs. task-oriented), all of which shape the overall user experience with chatbots (De Cicco et al., 2020; Kuhail et al., 2022).

Recent studies in consumer behavior and interactive technology underscore the critical role of interaction in shaping users' favorable attitudes toward AI chatbots (Lin, 2024). Torres et al. (2024) provided empirical evidence that interactivity significantly influences both the cognitive and affective components of users' attitudes toward AI chatbots in e-commerce contexts. Their findings indicate that when users perceive chatbot interactions as highly interactive, contextually relevant, emotionally engaging, efficient, and seamless, they are more likely to develop favorable attitudes, which in turn reinforces the chatbot's role as a trusted and effective service channel. Similarly, Chakraborty et al. (2024) emphasized that responsiveness, accuracy, and user-friendliness are essential interaction components that enhance the overall user

experience and strengthen positive attitudes. When users perceive chatbot interactions as efficient, helpful, and seamless, they are more likely to evaluate the technology favorably. Ding and Najaf (2024) further highlighted that human-like interaction and conversational responsiveness enhance user expectations, thereby strengthening perceptions of chatbot reliability and attitudinal responses. Moreover, the overall interaction style, whether socially oriented or task-focused, was identified as a potential factor influencing users' trust and attitudes toward chatbots (De Cicco et al., 2020). These findings suggest that interaction quality and style extend beyond functional performance; they are intrinsically linked to users' psychological responses and affective evaluations, thereby shaping their overall perception and acceptance of AI chatbots.

In addition, interaction plays a fundamental role in influencing user satisfaction with AI chatbots. According to Orden-Mejía et al. (2023), factors such as informativeness, empathy, and interactivity positively influence user satisfaction when engaging with AI chatbots. Meanwhile, Al-Shafei (2024) reported that anthropomorphism and communication style in interaction are key drivers of satisfaction, indicating that user evaluations are influenced not only by system performance but also by the emotional resonance and social presence of chatbots. Jiang et al. (2022) provided empirical evidence that interaction significantly enhances user satisfaction, particularly through responsiveness and conversational tone. In practice, AI chatbots and virtual assistants deliver real-time, responsive communication that reduces perceived wait times and improves service efficiency, factors that contribute positively to user satisfaction, particularly in sectors such as banking, where immediacy is highly valued (Othayoth & Khanna, 2025). Furthermore, Cheng and Jiang (2020) asserted that effective interactions not only generate immediate user satisfaction but also promote continued use of AI chatbots. These findings collectively highlight that satisfaction with AI chatbots is significantly shaped by how interactive and engaging the system appears to users.

Accordingly, the hypotheses are formulated as follows:

H7. Interaction has a positive impact on Attitude toward AI chatbots.

H8. Interaction has a positive impact on Satisfaction with AI chatbots.

2.2.5. Impact of customization on attitude and satisfaction

Customization refers to the deliberate effort to proactively anticipate individual needs and deliver personalized services tailored to users' core interests and preferences, even without explicit requests (Alnaser et al., 2023; Misischia et al., 2022). In this context, chatbots can collect and store data from user interactions, such as conversations, purchase history, and behavioral activities (Misischia et al., 2022). This accumulated data allows for the generation of tailored content for each user, fostering a sense of personal relevance and direct engagement.

Customization is increasingly recognized as a critical determinant of user attitude in AI chatbot interactions. Research by Xia and Shannon (2024) demonstrated that customization in AI chatbot services significantly shapes users' attitudes and strengthens customer-brand relationships by enhancing perceptions of innovation. This suggests that when users perceive chatbot interactions as tailored to their individual needs and preferences, they are more likely to develop favorable attitudes due to a heightened sense of relevance and engagement. Supporting this perspective, Koziel et al. (2025) further emphasized that customization contributes to positive attitudinal outcomes by increasing perceived relevance, enjoyment, and user control within the technological environment. This finding aligns with motivation theory, which posits that intrinsic enjoyment, particularly that derived from engaging and personalized experiences, serves as a critical psychological driver in the formation of positive user attitudes. While prior studies have identified customization as an important determinant of attitudes across various sectors (Altarteer & Charissis, 2019; Kim, Chang, et al., 2023), its specific impact on post-adoption behavior in AI chatbot contexts remains underexplored, particularly among digitally native users in emerging

markets like Vietnam. By facilitating individualized and context-aware interactions, AI chatbots have the potential not only to enhance user experience but also to strengthen brand perception and attitudinal commitment.

Customization is identified as a key antecedent of user satisfaction in AI-enabled service environments, primarily through its ability to address individual user needs and enhance perceived service relevance (Kim & Kim, 2025). In digital banking, for instance, Alnaser et al. (2023) demonstrated that AI systems offering personalized interactions significantly improve satisfaction by aligning service delivery with users' expectations. Similarly, Chen, Le, and Florence (2021) found that when AI chatbots tailor responses to users' contextual preferences, perceptions of service quality and satisfaction notably increase, ultimately fostering continued usage. These findings support the theoretical premise that satisfaction arises not only from technical efficiency but also from the perceived fit between system behavior and user-specific expectations. Beyond functional aspects, customization also contributes to emotional engagement by enhancing users' sense of control, reducing perceived wait times, and increasing the relevance of information delivered (Giri et al., 2024; Lee & Chan, 2023; Park & Lee, 2023). This experiential alignment reinforces trust and satisfaction, which are central to long-term system adoption. Jiang et al. (2022) also revealed that chatbot services characterized by accurate, competent, customized, and credible communication can significantly reduce user uncertainty, thereby enhancing overall customer satisfaction. These findings collectively underscore that customization is a fundamental driver of user satisfaction and sustained engagement in AI-powered service environments, affirming its strategic importance in the design of personalized digital interactions. Despite growing recognition of its importance, existing research has largely focused on general service contexts or broader technology platforms, with relatively little attention to AI chatbot applications in emerging markets. This study addresses this gap by empirically examining how chatbot-driven customization affects post-adoption satisfaction among Vietnam's digitally native population, providing fresh insights into how personalized experiences influence continued engagement in AI-mediated customer service.

Accordingly, the hypotheses are formulated as follows:

H9. Customization has a positive impact on Attitude toward AI chatbots.

H10. Customization has a positive impact on Satisfaction with AI chatbots.

2.2.6. *Impacts of problem-solving on attitude and satisfaction*

Problem-solving in AI applications denotes the system's ability to autonomously handle user inquiries and requests in real time through AI, thus ensuring uninterrupted service delivery without reliance on time-bound human intervention (Shankar, 2018; Xu et al., 2020). By leveraging technologically advanced systems, including NLP and ML, AI chatbots can simulate human reasoning, interpret user intent, extract key information, and deliver accurate and coherent responses based on structured knowledge sources (Kaplan & Haenlein, 2018; Rane et al., 2024). As a result, this functionality empowers AI chatbots to serve as effective problem-solvers across diverse user scenarios, thereby shaping users' perceptions of the technology and its practical value.

Empirical studies highlight a strong correlation between the problem-solving capabilities of AI chatbots and user attitudes. When users perceive chatbots as competent in resolving their issues, they are more likely to form favorable attitudes, largely due to enhanced perceptions of system reliability and competence (Chi, 2024). Supporting this, Xia and Shannon (2024) argued that problem-solving is not merely a technical backend function but a pivotal factor influencing user attitude. Sereno III (2025) further demonstrated that AI tools such as ChatGPT influence students' attitudes toward educational technologies by improving their ability to solve problems. Beyond its direct effects on attitudes, problem-solving, along with analyzing and reflecting,

significantly enhances the perceived ease of use of digital technologies, which in turn positively influences attitudes toward digital learning (Scheel et al., 2022). Particularly, in academic or task-driven environments, chatbots' ability to resolve issues enhances users' sense of support and efficiency, thereby contributing significantly to attitude formation. Together, these findings highlight problem-solving as a critical functional driver in the attitudinal acceptance of AI-powered technologies.

Problem-solving is also considered a critical factor in shaping and enhancing user satisfaction in AI chatbot interactions. Hsiao and Chen (2021) identified problem-solving as one of the influential service quality dimensions in shaping satisfaction, emphasizing that the effectiveness with which AI chatbots address user concerns plays a decisive role in user continuance intention. Alnaser et al. (2023) also demonstrated that effective problem resolution fosters satisfaction by reducing uncertainty and improving user confidence in AI-driven services. Compared to more superficial system features, problem-solving plays a foundational role in meeting functional expectations, particularly in high-stakes or time-sensitive contexts like digital banking or customer service (Kondybayeva et al., 2024). Moreover, Rane et al. (2024) highlighted that when AI chatbots demonstrate proactive problem-solving, rather than simply reacting passively, thereby improving overall satisfaction and fostering trust between users and the system. These insights indicate that problem-solving not only enhances immediate user satisfaction but also serves as a pivotal antecedent of long-term user retention and system loyalty in AI chatbot interactions. Despite its established relevance, the role of problem-solving in AI chatbot contexts remains underexplored in emerging markets such as Vietnam, especially among Generation Z users who value both efficiency and immediacy. This study addresses this gap by empirically investigating how chatbot problem-solving affects user attitudes and satisfaction in this demographic.

Accordingly, the hypotheses are formulated as follows:

H11. Problem-solving has a positive impact on Attitude toward AI chatbots.

H12. Problem-solving has a positive impact on Satisfaction with AI chatbots.

2.2.7. *Impacts of perceived enjoyment on attitude and satisfaction*

Perceived enjoyment refers to the degree to which interacting with a specific information technology is intrinsically pleasurable, irrespective of its functional performance or utilitarian outcomes (Jo, 2022; Venkatesh & Davis, 2000). In the context of digital environments, particularly in chatbot interactions, perceived enjoyment is regarded as a critical indicator of user acceptance and positive experience with technology (Rese et al., 2020; Selamat & Windasari, 2021).

Empirical evidence underscores the role of perceived enjoyment as a crucial antecedent of users' attitudes toward AI chatbots. The intrinsically driven pleasure experienced during chatbot interaction plays a pivotal role in determining positive attitudes and fostering intention to use, rather than being solely influenced by functional benefits (Kasilingam, 2020). In line with this argument, De Cicco et al. (2020) demonstrated that perceived enjoyment serves as a key explanatory factor for user attitudes, meaning that the pleasurable experience during interaction significantly determines whether users perceive AI chatbots positively. Extending these insights, Kim et al. (2024) noted that perceived enjoyment contributes not only to users' attitudes toward AI chatbots in customer service but also to their usage intentions toward AI. Similarly, Khrouf et al. (2025) confirmed that perceived enjoyment enhances users' attitudes toward chatbots, which in turn reinforces continuance intention. These findings position perceived enjoyment as a core psychological factor in technology acceptance frameworks.

In addition to attitude formation, perceived enjoyment significantly influences user satisfaction with AI chatbots. When users perceive the interaction as enjoyable, it suggests that their experience not only meets but often exceeds expectations, thereby enhancing overall satisfaction (Huang et al., 2024; Park & Lee, 2023). In this context, perceived

enjoyment not only serves as an indicator of user satisfaction but also functions as a reinforcing factor in the user-system relationship, subsequently promoting continued usage intentions. Orden-Mejía et al. (2024) provided empirical support for this relationship, revealing that feelings of enjoyment during interaction with AI chatbots exert a significant positive influence on user satisfaction. Notably, in emotionally sensitive domains such as tourism, perceived enjoyment is identified as a key antecedent of satisfaction with AI chatbots in services (Pereira et al., 2021, 2022). These findings suggest that when users derive emotional satisfaction from engaging with AI chatbots, they are predisposed to form favorable perceptions of the system.

Accordingly, the hypotheses are formulated as follows:

H13. Perceived Enjoyment has a positive impact on Attitude toward AI chatbots.

H14. Perceived Enjoyment has a positive impact on Satisfaction with AI chatbots.

2.2.8. Impacts of attitude on continuance intention

Attitude is conceptualized as an individual's overall evaluation of performing a particular behavior, typically expressed through favorable or unfavorable feelings toward that behavior (Ajzen, 1991). Within the Theory of Planned Behavior (TPB), which builds upon the premises of the Theory of Reasoned Action (TRA), attitude is a foundational component that significantly influences behavioral intention (Ajzen, 1991). In the digital context, particularly regarding AI-powered tools such as chatbots, attitude emerged as a central psychological factor affecting users' willingness to continue engagement. Continuance intention, in this setting, denotes a user's ongoing intention to use a system beyond the initial adoption phase, shaped by factors such as perceived usefulness, satisfaction, and habitual interaction (Bhattacharjee, 2001). For AI chatbots, which are increasingly embedded in customer service, education, and daily digital interaction, continuance intention is especially crucial. Sustained usage depends not solely on functional performance, but on the extent to which users experience favorable emotional evaluations toward aspects such as consistency, usefulness, and interaction quality (Foroughi et al., 2024).

Recent investigations consistently highlight the crucial influence of user attitude on continuance intention. Gao (2025) identified attitude as a strong predictor of users' intention to continue engaging with AI technologies. In addition, Huang and Lee (2022) demonstrated that attitudes toward fintech chatbots can be shaped by perceived social cues, drawing on social response theory, which in turn stimulate socially driven behaviors that positively influence continuance intention. This suggests that socially engaging and emotionally responsive interactions promote the growth of favorable attitudes, which are instrumental in encouraging long-term usage. According to a recent study by Xu et al. (2024), in their study on AI painting tools, confirmed that a positive attitude is a robust predictor of continuance intention, reinforcing the role of both emotional and cognitive evaluations in sustaining users' engagement with intelligent systems. Similarly, Pasupuleti and Thiyyagura (2024) demonstrated that higher education students with favorable attitudes toward ChatGPT exhibited stronger intentions to integrate it into their academic routines, emphasizing the generalizability of this relationship across user groups and technological domains. These findings suggest that fostering a positive user attitude toward AI chatbots is essential for promoting long-term usage and user loyalty.

Accordingly, the hypothesis is formulated as follows:

H15. Attitude has a positive impact on Continuance Intention toward AI chatbots.

2.2.9. Impacts of satisfaction on continuance intention

Satisfaction represents a user's emotional response, either positive or negative, to a consumed product or service, shaped by what they

respond to and when the response occurs (Giese & Cote, 2000). Within the framework of the Expectation-Confirmation Model (ECM), originally introduced by Oliver (1980), satisfaction arises when users recognize that the system's actual outcomes meet or exceed their initial anticipations. This theoretical model has been widely utilized to examine user acceptance and continuance behavior across various technology domains, including AI-driven interfaces such as chatbots (Ashfaq et al., 2020).

Within the context of AI chatbot usage, satisfaction is widely recognized as a critical determinant of continuance intention. For instance, Esmaeilzadeh et al. (2025) demonstrated in health care services that user satisfaction exerts a significant influence on continuance intention, indicating that overall contentment with chatbot interactions strongly shapes users' willingness to maintain long-term engagement with such services. Moreover, Hsiao and Chen (2021) found that satisfaction exerted the strongest direct influence on users' continuance intention to use AI customer service chatbots in the context of food-ordering services. This indicates that users' overall contentment with their chatbot experience plays a pivotal role in determining their willingness to continue using such services. Similarly, Li et al. (2021) provided empirical support for a notable and positive correlation between satisfaction and continuance intention, showing that users whose expectations are fully met or even surpassed are more likely to repeatedly engage with chatbot services. Furthermore, Alvionita (2024) revealed that in the context of chatbot adoption, user satisfaction positively influences continuance intention, suggesting that a satisfying interaction with chatbots can lead to repeated usage and foster stronger emotional ties with the service provider. These findings underscore that satisfaction is not a fleeting response, but rather a fundamental driver of customer retention and behavioral loyalty in AI-powered service environments.

Accordingly, the hypothesis is formulated as follows:

H16. Satisfaction has a positive impact on Continuance Intention toward AI chatbots.

2.2.10. Research framework

This research proposes a comprehensive framework to examine how both system-related and marketing-related attributes of AI chatbots influence users' continuance intention, with attitude and satisfaction serving as central mediating variables. Drawing upon the ISSM, the framework incorporates key system-related factors, system quality, information quality, and service quality as foundational determinants of user experience. Additionally, it integrates marketing-oriented elements specific to chatbot interactions, including interaction, customization, problem-solving, and perceived enjoyment. These seven antecedents are hypothesized to affect users' attitude toward and satisfaction with the AI chatbots, which, in turn, are expected to shape their continuance intention. The conceptual model representing these hypothesized relationships is illustrated in Fig. 1.

3. Methodology

3.1. Questionnaire design

This research employed a quantitative research approach, utilizing a structured questionnaire to collect empirical data on the adoption of AI chatbots within customer service environments. The questionnaire comprised three main sections. The first section outlined the primary objective of this study, assured respondents of the confidentiality and academic use of their responses, and included screening questions to confirm prior experience with AI chatbots. The second section collected demographic information, such as gender, age, occupation, and average monthly income, to enable subgroup analysis and control for potential demographic effects. The third section included the 47 measurement items aligned with the core constructs of this study.

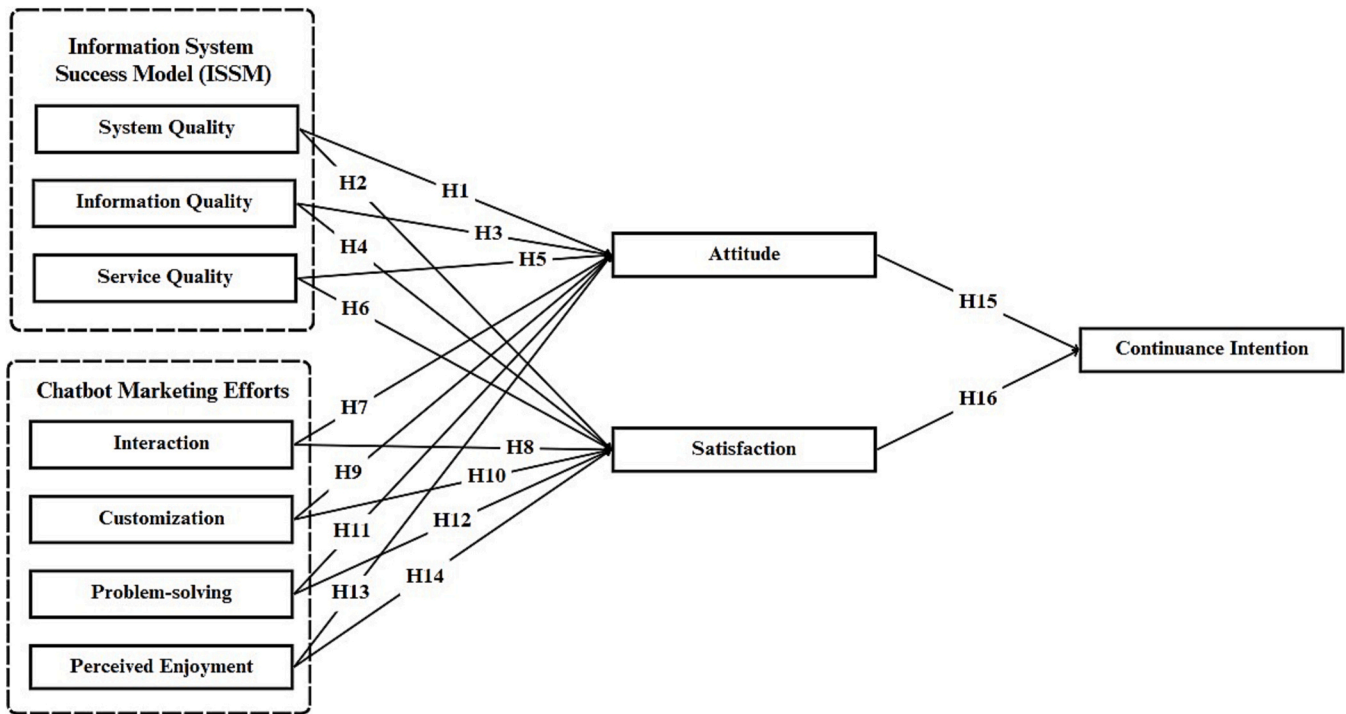


Fig. 1. Research framework.

The measurement items were derived from established and previously validated scales to ensure both content validity and relevance to the research context. The final instrument consisted of 47 items across 10 key constructs: System Quality (5 items), Information Quality (7 items), and Service Quality (5 items) were adapted from [Foroughi et al. \(2024\)](#); Interaction (4 items), Customization (4 items), and Problem-Solving (4 items) from [Chung et al. \(2018\)](#); Perceived Enjoyment (5 items) from [Ashfaq et al. \(2020\)](#) and [Pereira et al. \(2021\)](#); Attitude (3 items) from [De Cosmo et al. \(2021\)](#); Satisfaction (6 items) from [Singh and Singh \(2024\)](#); and Continuance Intention (4 items) from [Venkatesh et al. \(2012\)](#). Each construct was measured using multiple items to ensure comprehensive coverage of user perceptions and to enhance the reliability and validity of the results. To ensure the questionnaire's reliability and applicability within this study's context, it was reviewed and refined based on feedback from a marketing research expert before its finalization, thereby supporting robust and accurate data analysis.

All measurement items were evaluated using a five-point Likert-type scale, where 1 represents "strongly disagree" and 5 represents "strongly agree." This scale was chosen for its straightforwardness, clarity, and proven effectiveness in capturing attitudinal and behavioral data, as well as its suitability for statistical analysis techniques such as factor analysis and regression modeling. The overall questionnaire design was aimed at ensuring internal consistency and analytical rigor, thereby establishing a solid foundation for examining the factors influencing user engagement, satisfaction, and continuance intention toward AI chatbots in customer service contexts.

3.2. Data collection

For the data collection, this research utilized convenience sampling, a non-probability convenience sampling technique. This approach was chosen due to its practical benefits regarding accessibility, time efficiency, and cost-effectiveness. This sampling approach aligns with the recommendations of [Acharya et al. \(2013\)](#), who emphasized its appropriateness for exploratory research where generalizability is not the primary objective. In this study, the questionnaire was administered via both online and offline channels using a mixed-mode survey approach.

This method has been shown to effectively reduce nonresponse and measurement error bias, particularly when a high proportion of respondents participate online ([Mauz et al., 2018](#); [Sakshaug et al., 2018](#)). Online distribution was facilitated via Google Forms and shared through email and relevant online communities focused on chatbots and AI, including platforms such as Facebook and Zalo. Offline data collection was conducted at university campuses, offices, and public parks, targeting individuals with prior experience in using digital customer service tools.

To guarantee linguistic and cultural appropriateness, the questionnaire was originally established in English and subsequently translated into Vietnamese following the back-translation method as outlined by [Brislin \(1970\)](#). This process involved an initial translation into Vietnamese, followed by a reverse translation into English by an independent translator to ensure semantic equivalence and minimize potential biases related to language or cultural interpretation.

Ethical approval for this study was obtained from the Board of Directors at FPT Can Tho University, Vietnam (Approval No. 20250201.01). The research adhered strictly to ethical guidelines governing human subjects research, including obtaining fully informed agreement, ensuring uncoerced involvement, and ensuring the confidentiality of participants. Data collection was conducted between February 20 and March 20, 2025.

The required minimum sample size was determined following the guidelines of [Hair et al. \(1987\)](#), which suggest that in quantitative studies the sample should be at least five times the number of observed variables included in the model. Given the 49 observed variables in the model, a minimum of 245 valid responses was necessary. This study initially collected 623 responses through online and offline surveys. Following rigorous data quality screening, 562 valid responses were retained for analysis. Several exclusion criteria were applied to ensure data integrity and relevance. Firstly, participants without prior exposure to AI chatbot usage were excluded, as their input would not align with this study's context. Secondly, responses submitted within an implausibly short duration were removed, as such behavior, often referred to as satisficing, can indicate inattentive or careless responding ([Schneider et al., 2022](#)). Finally, only participants who fell within the defined

Generation Z cohort (individuals whose birth years range from 1995 to 2010, as defined by Francis & Hoefel, 2018) were included in the final sample to align with this study's target population.

3.3. Data analysis

To analyze the associations among user attitude, satisfaction, and continuance intention toward AI chatbots, this study employed Partial Least Squares Structural Equation Modeling (PLS-SEM). This method was considered suitable due to its capacity to support exploratory research endeavors, its strength in prediction-oriented analysis, and its capacity to represent intricate associations among multiple latent constructs in theory development contexts (Sarstedt et al., 2021). SmartPLS version 3 was used for all analyses, including a bootstrapping procedure with 5000 subsamples and two-tailed testing to estimate standard errors and assess statistical significance.

The analytical process was conducted in two stages. At the initial stage, the measurement model was evaluated to confirm the reliability and validity of the constructs. Internal consistency was assessed using Cronbach's Alpha and Composite Reliability (CR), while convergent validity was established via the Average Variance Extracted (AVE). Discriminant validity was assessed using both the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio, ensuring that constructs were conceptually distinct. Multicollinearity among predictor variables was also examined using the Latent Variable Correlations and Variance Inflation Factor (VIF), confirming that no variable exerted disproportionate influence on the model due to overlap with other predictors.

In the second stage, the structural model was evaluated to test the proposed hypotheses. The analysis included examining path coefficients (β), standard deviations, t-statistics, and *p*-values obtained through bootstrapping. These metrics allowed for an assessment of both the magnitude and significance of relationships among constructs. The model's explanatory power was determined using R-squared (R^2) values, which indicate the proportion of variance in the endogenous variables explained by the predictors. Predictive relevance was further examined using Stone-Geisser's Q^2 statistic via blindfolding procedures, providing an additional check on the model's capacity to forecast unobserved data.

To enhance the managerial relevance of the findings, an Importance-Performance Map Analysis (IPMA) was conducted, focusing on continuance intention as the key outcome variable. IPMA combines the importance of each construct (measured through total effects) with its performance (measured by average latent variable scores) to identify areas that are both critical to success and in need of improvement. This additional layer of analysis supports the formulation of actionable recommendations aimed at enhancing user retention and satisfaction with AI chatbots in customer service.

4. Results

4.1. Participant demographic information

Following the data screening process, a total of 562 valid responses were retained for analysis. In terms of gender distribution, 65 % of respondents identified as female ($n = 366$), while 35 % identified as male ($n = 196$), as presented in Table 1. The age distribution confirmed this study's focus on Generation Z, with 37.7 % of respondents aged between 18 and 22 years, 35.9 % aged 23 to 26 years, and 26.4 % aged 27 to 30 years. Regarding monthly income, the largest group (33.4 %) reported earning less than 5 million VND, followed by 28.3 % earning between 5 million and under 10 million VND. Smaller proportions fell within the income brackets of 10 to under 20 million VND (19.2 %), 20 to under 30 million VND (13.9 %), and over 30 million VND (5.2 %). These figures suggest a diverse economic representation among respondents, reflecting the broader Generation Z population in Vietnam.

Participants reported using AI chatbots across diverse sectors, with

Table 1
Demographic information.

Respondents' Information ($N = 562$)		Frequency	Percentage
Gender	Male	196	35 %
	Female	366	65 %
Age	18–22 years old	212	37.7 %
	23–26 years old	202	35.9 %
	27–30 years old	148	26.4 %
	Under 5 million VND	188	33.4 %
Monthly Income	From 5 - under 10 million VND	159	28.3 %
	From 10 - under 20 million VND	108	19.2 %
	From 20 - under 30 million VND	78	13.9 %
	Above 30 million VND	29	5.2 %
	Finance and Banking	137	24.4 %
	E-commerce (Online Shopping)	135	24.0 %
	Travel and Aviation	78	13.9 %
Usage Sectors	Healthcare and Medical Services	41	7.3 %
	Education and Learning	114	20.3 %
	Insurance	25	4.5 %
	Social Medias	137	24.4 %
	Others	28	5.0 %
	Product/Service Inquiries	150	26.7 %
	Technical Support or Troubleshooting	138	24.6 %
	Consultation on Pricing or Promotions	90	16.0 %
	Guidance on Product or Service Usage	131	23.3 %
	Appointment Scheduling or Purchase Requests	135	24.0 %
Usage Purposes	Others	30	5.3 %

the highest usage observed in social media and finance and banking (24.4 %), followed by e-commerce (24.0 %) and education and learning (20.3 %). These figures reflect the widespread, cross-sector adoption of chatbot technology. In terms of interaction purposes, the most common reasons included requesting information about products or services (26.7 %), supporting techniques or troubleshooting (24.6 %), and scheduling appointments or conducting transactions (24.0 %). Other uses involved guidance on product or service usage (23.3 %), consultation on pricing or promotions (16.0 %), and other purposes (5.3 %).

These demographic and usage insights, summarized in Table 1, provide a solid foundation for analyzing how factors such as system quality, information quality, service quality, interaction, customization, problem-solving, and perceived enjoyment influence user attitudes, satisfaction, and continuance intentions, particularly among Generation Z users in Vietnam.

4.2. Measurement model assessment

Examining the measurement model is a fundamental step in PLS-SEM to ensure that the constructs are both statistically reliable and conceptually valid before proceeding to structural analysis. This process involves evaluating construct reliability and convergent validity using indicators such as Cronbach's Alpha, Composite Reliability (CR), Outer Loadings, and Average Variance Extracted (AVE). In addition, discriminant validity is assessed through the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio to confirm that constructs are empirically distinct. Together, these evaluations establish the psychometric adequacy of the measurement model and support its suitability for subsequent structural analysis.

4.2.1. Construct reliability and validity

Reliable and valid measurement is widely recognized as the cornerstone of empirical research, particularly in PLS-SEM, where the accuracy of measurement directly influences the credibility of theoretical interpretations and the strength of structural relationships (Sarstedt et al., 2021). In PLS-SEM, establishing construct reliability and convergent validity is essential, as poorly measured constructs may

introduce systematic bias, inflate or deflate path coefficients, and compromise the model's overall explanatory power (Schuberth et al., 2016). Therefore, rigorous assessment of construct reliability and convergent validity is not merely a technical formality but a prerequisite for theoretical integrity and statistical robustness in PLS-SEM analysis (Roberts & Priest, 2006).

To assess internal consistency reliability, this study used both Cronbach's Alpha and Composite Reliability (CR). Cronbach's alpha is used as a metric to evaluate the internal consistency or reliability of items within a construct (Bujang et al., 2018). While Cronbach's Alpha values of 0.70–0.80 are deemed acceptable, values above 0.80 are generally considered acceptable, values above 0.80 indicate high reliability, particularly in confirmatory research contexts (Hulin et al., 2001). As presented in Table 2, Cronbach's Alpha values ranged from 0.863 to 0.946, exceeding the recommended threshold of 0.80, indicating strong internal consistency. Complementarily, Composite Reliability (CR) values ranged from 0.907 to 0.958, surpassing the 0.70 benchmark recommended for confirmatory (Hair et al., 2013), further supporting the reliability of the constructs.

In terms of indicator reliability, outer loadings reflect the strength of the association between indicators and their latent constructs, serving as a key measure of indicator reliability and construct validity in reflective models (Hair Jr. et al., 2016). According to Sarstedt et al. (2021), outer loadings should exceed 0.708 to ensure sufficient indicator reliability. In this study, all outer loadings met or exceeded this threshold, ranging from 0.798 (PE3) to 0.933 (ATT1), indicating strong associations between the indicators and their respective constructs.

Convergent validity was assessed using the Average Variance Extracted (AVE), which indicates the portion of variance explained by a construct compared to the variance due to measurement error (Santos & Cirillo, 2021). According to Hair Jr. et al. (2009), it is commonly expected that the AVE for each construct exceeds 0.5 to meet the convergent validity requirement. In this study, AVE values ranged from 0.681 to 0.860, exceeding the 0.50 criterion, thus confirming that each construct accounts for a significant portion of variance in its indicators. Overall, the measurement model exhibits strong internal consistency, robust indicator reliability, and satisfactory convergent validity. These results provide a solid empirical foundation for testing the structural relationships within the model, particularly in the context of the AI chatbots' usage in digital customer service.

4.2.2. Discriminant validity

Discriminant validity is a critical aspect of measurement model assessment, ensuring that each construct is conceptually and statistically distinct from others in the model (Fornell & Larcker, 1981; Hair et al., 2018). Establishing discriminant validity confirms that the constructs reflect distinct theoretical concepts rather than overlapping aspects of the same phenomenon. A failure to confirm discriminant validity can lead to ambiguity in interpreting the structural model and compromise the credibility of empirical findings (Henseler et al., 2014). In this study, discriminant validity was examined using two mutually supportive techniques: the Heterotrait-Monotrait ratio of correlations (HTMT) and the Fornell-Larcker criterion, both of which are widely accepted and methodologically rigorous (Paudel, 2023).

The HTMT ratio, regarded as a more sensitive and reliable method than traditional criteria, evaluates discriminant validity by contrasting the average correlations between different constructs with those within the same construct, thereby providing a robust estimate of inter-construct distinctiveness (Henseler et al., 2014). Thresholds for acceptable HTMT values vary depending on the context: a strict threshold of 0.85 is suggested for models requiring high construct distinctiveness, while a threshold of 0.90 may be acceptable in exploratory or behavioral studies (Rasoolimanesh, 2022; Sarstedt et al., 2021). As shown in Table 3, HTMT values ranged from 0.445 to 0.731, while several values approached the threshold of 0.70, such as those between Continuance Intention (CI) and Satisfaction (SAT) at 0.711, and

Table 2

Convergent validity.

Constructs' indicators	Outer loadings	Cronbach's alpha (α)	Composite reliability (CR)	Average variance extracted (AVE)
System quality (SQ)		0.895	0.923	0.704
SQ1. "AI chatbots are reliable."	0.851			
SQ2. "AI chatbots are quick in response."	0.839			
SQ3. "AI chatbots understand my queries well."	0.847			
SQ4. "AI chatbots memorize my questions, needs, or preferences."	0.853			
SQ5. "AI chatbots respond in a natural and human-like manner."	0.806			
Information quality (IQ)		0.922	0.937	0.681
IQ1. "AI chatbots provide sufficient information."	0.805			
IQ2. "Through AI chatbots, I get the information I need on time."	0.813			
IQ3. "Information provided by AI chatbots is in a useful format."	0.815			
IQ4. "Information provided by AI chatbots is clear."	0.828			
IQ5. "Information provided by AI chatbots is accurate."	0.878			
IQ6. "Information provided by AI chatbots is up-to-date."	0.830			
IQ7. "Information provided by AI chatbots is reliable."	0.803			
Service quality (SEQ)		0.902	0.927	0.718
SEQ1. "AI chatbots have a modern-looking interface."	0.857			
SEQ2. "AI chatbots provide the right solution to my request."	0.803			
SEQ3. "AI chatbots have visually appealing materials."	0.860			
SEQ4. "AI chatbots give me personalized attention."	0.885			
SEQ5. "AI chatbots have an excellent interface to communicate my needs."	0.829			
Interaction (IN)		0.881	0.918	0.737
IN1. "AI chatbots have the knowledge to effectively answer customers' questions."	0.870			

(continued on next page)

Table 2 (continued)

Constructs' indicators	Outer loadings	Cronbach's alpha (α)	Composite reliability (CR)	Average variance extracted (AVE)
IN2. "AI chatbots are always available to respond to users' requests without delay."	0.841			
IN3. "AI chatbots provide personalized attention to users."	0.863			
IN4. "AI chatbots consistently interact with users."	0.859			
Customization (CU)		0.872	0.913	0.723
CU1. "AI chatbots offer personalized services and support that I couldn't find with others."	0.840			
CU2. "If I switched to another chatbot, the chatbot's level of customization and service would not match what I currently experience."	0.845			
CU3. "I feel that interacting with these AI chatbots and using their services effectively meets my personal needs."	0.855			
CU4. "AI chatbots provide information about products and services tailored to my preferences."	0.861			
Problem-solving (PS)		0.863	0.907	0.709
PS1. "AI chatbots efficiently assist with handling returns and exchanges."	0.846			
PS2. "When a customer has a problem, AI chatbots show a sincere interest in solving it."	0.849			
PS3. "AI chatbots are capable of addressing customer complaints directly and immediately."	0.845			
PS4. "I have confidence that the AI chatbots can complete tasks."	0.828			
Perceived enjoyment (PE)		0.887	0.917	0.689
PE1. "I like to chat with AI chatbots."	0.849			
PE2. "It is fun and enjoyable to share a conversation with AI chatbots."	0.857			
PE3. "The conversation with AI chatbots is exciting."	0.798			
PE4. "I like to choose products more if they are	0.825			

Table 2 (continued)

Constructs' indicators	Outer loadings	Cronbach's alpha (α)	Composite reliability (CR)	Average variance extracted (AVE)
recommended by AI chatbots than if I want them myself."				
PE5. "I was absorbed in the conversation with AI chatbots."	0.819			
Attitude (ATT)		0.919	0.949	0.860
ATT1. "Using AI chatbots for customer service seems like a good idea."	0.933			
ATT2. "AI chatbots make interacting with customer service more interesting."	0.923			
ATT3. "Using AI chatbots seems fun."	0.927			
Satisfaction (SAT)		0.946	0.958	0.821
SAT1. "AI chatbots meet my expectations for quality customer service."	0.892			
SAT2. "AI chatbots' interactions leave me with a positive impression of the company."	0.908			
SAT3. "AI chatbots contribute to my overall satisfaction as a customer."	0.906			
SAT4. "I feel valued as a customer when interacting with AI chatbots."	0.909			
SAT5. "Overall, I am satisfied with my experiences with AI chatbots."	0.916			
Continuance intention (CI)		0.862	0.906	0.707
CI1. "I intend to continue using the AI chatbots rather than discontinue their use."	0.835			
CI2. "I intend to continue using the AI chatbots rather than use any alternative means (such as traditional customer service methods)."	0.847			
CI3. "If I could, I would prefer to keep using the AI chatbots."	0.829			
CI4. "I plan to continue to use AI chatbots frequently."	0.852			

Continuance Intention (CI) and Service Quality (SEQ) at 0.731; all remained below the conservative cut-off value of 0.85. These results affirm that the constructs are empirically separate and satisfy the HTMT criterion for discriminant validity.

To reinforce these findings, the Fornell-Larcker criterion was also applied as a supplementary diagnostic. This approach evaluates

Table 3
HTMT (Heterotrait-Monotrait ratio).

	ATT	CI	CU	IN	IQ	PE	PS	SAT	SEQ	SQ
ATT										
CI	0.733									
CU	0.560	0.703								
IN	0.509	0.646	0.498							
IQ	0.514	0.710	0.547	0.469						
PE	0.500	0.624	0.502	0.554	0.451					
PS	0.461	0.621	0.480	0.497	0.509	0.487				
SAT	0.535	0.708	0.453	0.444	0.535	0.465	0.464			
SEQ	0.524	0.731	0.543	0.494	0.578	0.452	0.456	0.538		
SQ	0.517	0.691	0.456	0.469	0.532	0.412	0.450	0.542	0.518	

Note: ATT = Attitude; CI = Continuance Intention; CU = Customization; IN = Interaction; IQ = Information Quality; PE = Perceived Enjoyment; PS = Problem-Solving; SAT = Satisfaction; SEQ = Service Quality; SQ = System Quality.

discriminant validity by comparing the square root of the Average Variance Extracted (AVE) for each construct to its correlations with all other constructs. A construct is considered to possess sufficient discriminant validity when the square root of the AVE for a construct is greater than its correlations with any other constructs in the model (Fornell & Larcker, 1981). Results of the Fornell-Larcker assessment in Table 4 indicated that, for all constructs, this condition was consistently met. Therefore, the AVE-based evaluation further supports the notion that the constructs are statistically separable and non-redundant. Together, the results of the HTMT analysis and the Fornell-Larcker criterion provide strong and convergent evidence for the discriminant validity of the measurement model. These findings confirm the psychometric robustness of the constructs and support their suitability for investigating structural relationships in the context of the AI chatbots application in digital customer service.

4.3. Structural model assessment

4.3.1. Latent variable correlations analysis

Latent variable correlations represent the bivariate linear relationships between core constructs in the structural model, providing foundational insight complementary to the PLS-SEM results (Wong, 2013). This preliminary analysis helps assess construct distinctiveness and potential multicollinearity while revealing the strength and direction of associations within the model under study. The results from the correlation matrix shown in Table 5 demonstrated predominantly moderate to strong positive correlations among all latent variables, with values roughly ranging between 0.369 (between PE and SQ) and 0.653 (between ATT and CI). The strongest association was observed between Attitude (ATT) and Continuance Intention (CI), indicating a robust positive relationship between users' attitudes and their intention to continue using AI chatbots. System Quality (SQ), Information Quality (IQ), and Service Quality (SEQ) exhibit substantial positive correlations with Attitude (ATT), Satisfaction (SAT), and Continuance Intention (CI), supporting their crucial role in influencing user perceptions and

behavioral intentions. Experiential constructs such as Perceived Enjoyment (PE), Interaction (IN), and Customization (CU) also show meaningful positive relationships with Attitude (ATT) and Satisfaction (SAT), indicating their contribution to affective user responses. Overall, the latent variable correlations affirm the harmonious interrelations among constructs that underpin the structural model, providing empirical support for the theoretical framework and subsequent hypothesis testing.

4.3.2. Assessment of common method bias (CMB)

Assessing inner Variance Inflation Factor (VIF) values is a necessary step in structural model evaluation to identify potential multicollinearity and Common Method Bias (CMB), these issues that can compromise the accuracy of path estimates in PLS-SEM (Kock, 2015; Sarstedt et al., 2021). This diagnostic is especially important in studies using self-reported data, where CMB may arise when respondents give similar answers across constructs due to overlapping items or social desirability, potentially inflating inter-construct correlations and reflecting method effects rather than true relationships (Podsakoff et al., 2003).

According to Hair et al. (2011), VIF values below 5.0 are commonly accepted as indicating an absence of critical collinearity, while Kock (2015) proposes a more conservative threshold of 3.3, above which common method bias may be suspected. As summarized in Table 6, all inner VIF values ranged from 1.332 to 1.751. Specifically, Information Quality (IQ) recorded the highest VIF at 1.751, while Attitude (ATT) and Satisfaction (SAT) exhibited the lowest at 1.332. These values are well below both the 5.0 and 3.3 thresholds, indicating the absence of problematic multicollinearity or common method bias. These results provide compelling empirical evidence that the structural model is not adversely affected by measurement artifacts. Consequently, the estimated path coefficients can be interpreted as statistically reliable and theoretically valid, supporting the robustness of the relationships among constructs in the proposed model.

Table 4
Fornell-Larcker criterion.

	ATT	CI	CU	IN	IQ	PE	PS	SAT	SEQ	SQ
ATT	0.928									
CI	0.653	0.841								
CU	0.502	0.609	0.850							
IN	0.459	0.563	0.437	0.859						
IQ	0.474	0.634	0.491	0.423	0.825					
PE	0.454	0.549	0.445	0.490	0.408	0.830				
PS	0.410	0.537	0.417	0.433	0.456	0.426	0.842			
SAT	0.499	0.640	0.412	0.405	0.500	0.429	0.420	0.906		
SEQ	0.477	0.646	0.482	0.441	0.528	0.405	0.403	0.498	0.847	
SQ	0.469	0.607	0.404	0.416	0.484	0.369	0.396	0.499	0.466	0.839

Note: ATT = Attitude; CI = Continuance Intention; CU = Customization; IN = Interaction; IQ = Information Quality; PE = Perceived Enjoyment; PS = Problem-Solving; SAT = Satisfaction; SEQ = Service Quality; SQ = System Quality.

Table 5

Latent variable correlations.

	ATT	CI	CU	IN	IQ	PE	PS	SAT	SEQ	SQ
ATT	1.000	0.653	0.502	0.459	0.474	0.454	0.410	0.499	0.477	0.469
CI	0.653	1.000	0.609	0.563	0.634	0.549	0.537	0.640	0.646	0.607
CU	0.502	0.609	1.000	0.437	0.491	0.445	0.417	0.412	0.482	0.404
IN	0.459	0.563	0.437	1.000	0.423	0.490	0.433	0.405	0.441	0.416
IQ	0.474	0.634	0.491	0.423	1.000	0.409	0.456	0.500	0.528	0.484
PE	0.454	0.549	0.445	0.490	0.409	1.000	0.426	0.429	0.405	0.369
PS	0.410	0.537	0.417	0.433	0.456	0.426	1.000	0.420	0.403	0.396
SAT	0.499	0.640	0.412	0.405	0.500	0.429	0.420	1.000	0.498	0.499
SEQ	0.477	0.646	0.482	0.441	0.528	0.405	0.403	0.498	1.000	0.466
SQ	0.469	0.607	0.404	0.416	0.484	0.369	0.396	0.499	0.466	1.000

Note: ATT = Attitude; CI = Continuance Intention; CU = Customization; IN = Interaction; IQ = Information Quality; PE = Perceived Enjoyment; PS = Problem-Solving; SAT = Satisfaction; SEQ = Service Quality; SQ = System Quality.

Table 6

Results of testing hypotheses in the theoretical framework.

H	Structural path	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics	P-values	VIF	Result
H1	SQ → ATT	0.160	0.161	0.039	4.093	0.000	1.523	Accepted
H2	SQ → SAT	0.214	0.214	0.041	5.239	0.000	1.523	Accepted
H3	IQ → ATT	0.105	0.106	0.045	2.320	0.021	1.751	Accepted
H4	IQ → SAT	0.166	0.167	0.044	3.784	0.000	1.751	Accepted
H5	SEQ → ATT	0.126	0.128	0.046	2.736	0.006	1.682	Accepted
H6	SEQ → SAT	0.181	0.181	0.044	4.138	0.000	1.682	Accepted
H7	IN → ATT	0.118	0.116	0.042	2.780	0.006	1.599	Accepted
H8	IN → SAT	0.044	0.045	0.044	1.001	0.317	1.599	Rejected
H9	CU → ATT	0.188	0.188	0.042	4.514	0.000	1.616	Accepted
H10	CU → SAT	0.041	0.039	0.046	0.879	0.380	1.616	Rejected
H11	PS → ATT	0.062	0.064	0.040	1.540	0.124	1.504	Rejected
H12	PS → SAT	0.096	0.098	0.041	2.318	0.021	1.504	Accepted
H13	PE → ATT	0.134	0.134	0.040	3.326	0.001	1.543	Accepted
H14	PE → SAT	0.127	0.126	0.043	2.998	0.003	1.543	Accepted
H15	ATT → CI	0.445	0.447	0.026	16.781	0.000	1.332	Accepted
H16	SAT → CI	0.419	0.419	0.027	15.712	0.000	1.332	Accepted

Note: SQ = System Quality; IQ = Information Quality; SEQ = Service Quality; IN = Interaction; CU = Customization; PS = Problem-Solving; PE = Perceived Enjoyment; ATT = Attitude; SAT = Satisfaction; CI = Continuance Intention.

4.3.3. Examination of research hypotheses

The results of hypothesis testing, as detailed in Table 6 and illustrated in Fig. 2, offer a comprehensive understanding of the factors influencing user Attitude (ATT), Satisfaction (SAT), and Continuance Intention (CI) in the context of AI chatbots customer service. The findings provide robust empirical support for most hypothesized relationships, underscoring the importance of various quality dimensions and experiential features.

The analysis confirms that System Quality (SQ) significantly influences Attitude (ATT) (H1: $O = 0.160$, $t = 4.093$, $p < 0.001$) and Satisfaction (SAT) (H2: $O = 0.214$, $t = 5.239$, $p < 0.001$), highlighting the role of technical proficiency and reliability of AI chatbots in shaping favorable user perceptions and experiences. Information Quality (IQ) also has significant positive effects on Attitude (ATT) (H3: $O = 0.105$, $t = 2.320$, $p = 0.021$) and Satisfaction (SAT) (H4: $O = 0.166$, $t = 3.784$, $p < 0.001$), indicating that accurate and relevant information delivered by the AI chatbots enhances both cognitive and emotional responses to chatbots.

Additionally, the findings reveal that Service Quality (SEQ) significantly contributes to both Attitude (ATT) (H5: $O = 0.126$, $t = 2.736$, $p = 0.006$) and Satisfaction (SAT) (H6: $O = 0.181$, $t = 4.138$, $p < 0.001$), underscoring the necessity for AI chatbots to provide precise and pertinent information to enhance positive attitude and foster user satisfaction. In contrast, Interaction (IN) positively affects Attitude (ATT) (H7: $O = 0.118$, $t = 2.780$, $p = 0.006$) but not Satisfaction (SAT) (H8: $O = 0.044$, $t = 1.001$, $p = 0.317$). This suggests that while interactive experiences foster more favorable evaluations, they may not directly increase satisfaction. Furthermore, Customization (CU) shows a strong positive effect Attitude (ATT) (H9: $O = 0.188$, $t = 4.514$, $p < 0.001$), suggesting that AI chatbots offering personalized experiences

are more likely to be viewed positively; though it does not significantly impact Satisfaction (SAT) (H10: $O = 0.041$, $t = 0.879$, $p = 0.380$).

In contrast, Problem-Solving (PS) significantly enhances Satisfaction (SAT) (H12: $O = 0.096$, $t = 2.318$, $p = 0.021$), indicating that chatbots capable of effectively addressing user concerns contribute positively to satisfaction. However, its effect on Attitude (ATT) is not significant (H11: $O = 0.062$, $t = 1.540$, $p = 0.124$), suggesting that its influence is more functional rather than shaping users' affective evaluations. These findings imply that while problem-solving enhances satisfaction, its impact on attitude may be mediated through other variables such as trust or perceived usefulness, rather than directly shaping users' overall perception of chatbots. The findings further indicate that Perceived Enjoyment (PE) positively contributes to user Attitude (ATT) (H13: $O = 0.134$, $t = 3.326$, $p = 0.001$), emphasizing the need to design AI chatbots that deliver enjoyable and intrinsically motivating experiences. Similarly, Perceived Enjoyment (PE) positively impacts Satisfaction (SAT) (H14: $O = 0.127$, $t = 2.998$, $p = 0.003$), highlighting the significance of creating enjoyable user experiences to elevate satisfaction levels. Furthermore, the analysis consistently demonstrates that both Attitude (AT) (H15: $O = 0.445$, $t = 16.781$, $p < 0.001$) and Satisfaction (SAT) (H16: $O = 0.419$, $t = 15.712$, $p < 0.001$) significantly affect Continuance Intention (CI), underscoring their crucial roles as key drivers of sustained engagement with AI chatbots.

The Original Sample (known as coefficients) values in Table 6 offer valuable insights into the relative strength and direction of each relationship within the model. The strong positive coefficients for H15 ($O = 0.445$) and H16 ($O = 0.419$) underscore the central role of attitude and satisfaction in influencing users' intentions to continue using AI chatbots. Among the remaining supported hypotheses (H1 through H14), with coefficients ranging from 0.041 to 0.214, it becomes apparent that

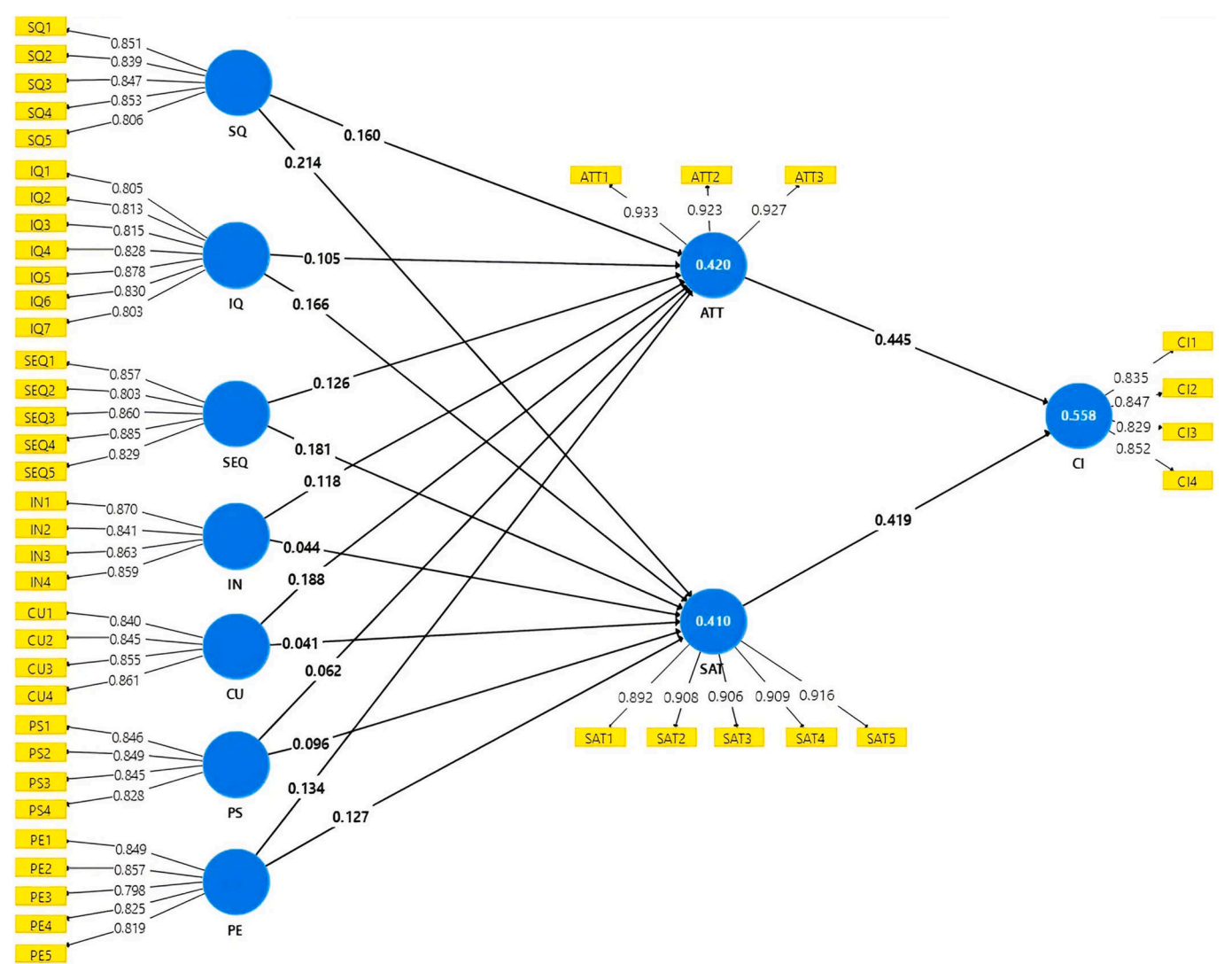


Fig. 2. PLS-SEM results (structural model).

these constructs contribute to building positive attitude and satisfaction, serving as key antecedents to continuance intention. Specifically, **H2** ($O = 0.214$) and **H9** ($O = 0.188$) exhibit the strongest effects among these antecedents, highlighting the importance of system quality in driving satisfaction and customization in shaping positive attitudes.

4.3.4. Model performance metrics

The effectiveness of the structural model is assessed based on its explanatory power and predictive relevance, as reflected by the indicators R^2 and Q^2 . The R^2 values represent the proportion of variance in an endogenous construct explained by its exogenous variables within the structural model (Sarstedt et al., 2021). In contrast, the Q^2 values, derived through the blindfolding procedure, evaluate the model's out-of-sample predictive relevance. A Q^2 value exceeding zero indicates that the model has predictive power beyond simple parameter estimation (Henseler et al., 2009).

According to thresholds suggested by Hair et al. (2011), R^2 values of 0.25, 0.50, and 0.75 reflect weak, moderate, and substantial explanatory power, respectively. In this study, as presented in Table 7, the R^2 for Attitude (ATT) is 0.420 and for Satisfaction (SAT) is 0.410, both indicating moderate explanatory power. Continuance Intention (CI) shows a higher R^2 of 0.558, suggesting a moderate-to-substantial degree of explanatory strength. These results suggest that the model explains 55.8 % of the variance in CI, 42.0 % in ATT, and 41.0 % in SAT, confirming

Table 7
Model performance metrics.

Constructs	R^2	Q^2
ATT	0.420	0.353
SAT	0.410	0.332
CI	0.558	0.391

Note: ATT = Attitude; SAT = Satisfaction; CI = Continuance Intention.

the structural model's effectiveness in accounting for user responses to AI chatbot interactions.

Regarding predictive relevance, Sarstedt et al. (2014) propose that Q^2 values of 0.02, 0.15, and 0.35 represent small, medium, and large predictive relevance, respectively. The Q^2 values obtained in this study were Attitude (ATT: 0.353), Satisfaction (SAT: 0.332), and Continuance Intention (CI: 0.391), as also reported in Table 7. These results suggest large predictive relevance for both Attitude (ATT) and Continuance Intention (CI), while Satisfaction (SAT) demonstrates high-medium predictive accuracy, falling just below the threshold for large relevance. Notably, Continuance Intention (CI) recorded the highest Q^2 value, indicating that the model can predict approximately 39 % of the variance in users' intention to continue using AI chatbots, thereby supporting the model's strong predictive validity.

Overall, these performance metrics indicate that the model not only

explains a meaningful proportion of variance in the key outcome variables but also exhibits strong predictive capability. This underscores the robustness and practical utility of the model in capturing user responses to AI chatbots' applications within digital customer service environments.

4.3.5. Importance-performance map analysis (IPMA)

To further complement the structural model assessment, the Importance-Performance Map Analysis (IPMA) was conducted, integrating the total effects (importance) of the constructs with their average latent variable scores (performance). This analysis identifies priority areas for managerial focus to enhance Continuance Intention (CI). According to Ringle and Sarstedt (2016), IPMA extends traditional PLS-SEM by highlighting constructs that not only have a strong influence on the target construct but also present opportunities for performance improvement.

As shown in Fig. 3, Satisfaction (SA) exhibits the highest importance (0.397) and performance (62.607), indicating that it is both a critical driver of continuance intention and is currently performing well. Attitude (ATT) follows closely, with an importance value of 0.382 and a performance score of 59.605, highlighting its essential role in promoting continuance intention while showing moderate performance. System Quality (SQ) demonstrates relatively high importance (0.157) and strong performance (60.826), reflecting its significant contribution to continuance intention and a satisfactory current state. In contrast, constructs such as Service Quality (SEQ) and Information Quality (IQ) show moderate importance (0.119 and 0.116, respectively) but comparatively lower performance scores (58.670 and 59.142). These areas indicate potential for improvement to maximize their impact on continuance intention. Perceived Enjoyment (PE) and Problem-Solving (PS) display lower importance values (0.111 and 0.061) with moderate performance scores (58.231 and 59.521), suggesting they contribute less strongly to continuance intention but still offer areas for potential enhancement. Overall, the IPMA results provide actionable insights by prioritizing constructs that combine high importance with lower performance. This approach guides efforts to optimize AI chatbots in customer service and improve their effectiveness in the digital context.

5. Discussion

This study investigated the influence of core dimensions from the ISSM, namely System Quality (SQ), Information Quality (IQ), and Service Quality (SEQ), in conjunction with chatbot marketing effort factors, including Interaction (IN), Customization (CU), Problem-Solving (PS), and Perceived Enjoyment (PE), on user Attitude (ATT), Satisfaction (SAT), and Continuance Intention (CI) toward AI chatbots usage in Vietnam. Particular attention was given to Generation Z, a tech-savvy

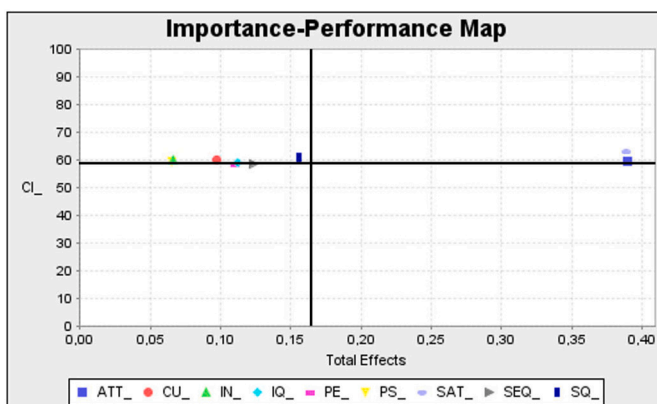


Fig. 3. Importance and performance map (priority map) of IPMA for continuance intention.

demographic that increasingly interacts with AI-driven services. The model was empirically tested using PLS-SEM, and the results provided support for the majority of the proposed hypotheses, with the exception of H8, H10, and H11, which were not confirmed.

5.1. Impacts of system quality on attitude and satisfaction

The results demonstrate that system quality exerts a significant positive impact on users' attitudes toward AI chatbots, thereby lending support to Hypothesis H1. This suggests that when the system is reliable, responsive, and adaptable to user needs, it fosters more favorable attitudes among users. The functional ease and seamless performance of the chatbots contribute to a positive evaluation, as users perceive the system to be dependable and aligned with their expectations. This outcome is consistent with prior research highlighting the pivotal role of system quality in shaping user acceptance. For instance, Rahi et al. (2025) suggested that system quality significantly influenced users' attitudes toward adopting service robots among Italian consumers. However, Magno and Dossena (2022) concluded that system quality has no significant relevance to the formation of customer attitudes or satisfaction toward chatbots in the brand relationship context. This inconsistency may be attributed to contextual factors such as national culture, economic development, and user demographics. As a developing country, Vietnam is still undergoing digital transformation, and users from Generation Z, who are highly familiar with digital technology, may place greater emphasis on technical efficiency and functional responsiveness when evaluating emerging technologies like AI chatbots. In this context, the fundamental performance of the system may carry more weight in user attitudes compared to digitally advanced environments, where users may prioritize relational or emotional aspects of the interaction. By confirming this relationship in the setting of AI-powered customer service, this research extends the generalizability of these findings and highlights the centrality of system quality in fostering positive user attitudes in AI interactions.

The findings also support Hypothesis H2, confirming that system quality exerts a notably positive impact on user satisfaction with AI chatbots in customer service. Satisfaction arises when the system operates consistently, provides timely responses, and handles user inquiries effectively. Users tend to feel content with the experience when technical issues are minimal, and the system reliably meets their service expectations. This result aligns with prior research emphasizing the role of system quality in driving satisfaction. For instance, Kim and Kim (2025) found that higher system quality significantly improved perceived quality of AI-generated recommendations, which subsequently enhanced user satisfaction. Similarly, Zainol et al. (2023) reported that specific components of system quality, namely reliability and availability, significantly influence customer satisfaction with banking chatbot services. In the AI-powered customer service context, Elayat and Elalfy (2025) also indicated that chatbot system compliance and responsiveness significantly improved satisfaction, particularly among younger users engaged in AI-powered customer service. These findings are especially relevant to the Vietnamese Generation Z context. As a highly connected and tech-savvy cohort, this generation expects immediate support and seamless interactions. They are particularly sensitive to delays, misunderstanding of intent, or system inflexibility. When the system fulfills functional expectations, such as delivering real-time feedback, offering accurate answers, and smoothly navigating queries, it strengthens the user's emotional response, contributing to the development of user satisfaction. In contrast to users in more digitally mature markets, where system reliability is often taken for granted, Vietnamese Generation Z users may view system quality as a key performance indicator for emerging technologies like AI chatbots.

5.2. Impacts of information quality on attitude and satisfaction

The findings of this research indicate that information quality exerts a significant positive influence on users' attitudes toward AI chatbots within customer service contexts, thus confirming Hypothesis H3. This suggests that when AI chatbots provide accurate, timely, clear, and well-organized information, users are more inclined to form favorable attitudes toward their use. These findings are in line with prior research by Kim, Li, and Choi (2023), who found that high levels of information quality significantly contribute to the formation of positive user attitudes toward chatbot services in e-commerce. Similarly, Kwangsawad and Jattamart (2022) reported a robust positive relationship between perceived information quality and user attitudes toward community-based customer service chatbots in Thailand. The consistency of these findings across studies highlights the critical role of information clarity and credibility in shaping user attitudes toward AI technologies. In the context of this study, which focuses on the Vietnamese Generation Z, a highly connected and digitally literate cohort, these findings are especially relevant. Generation Z users are quick to evaluate the usefulness and credibility of digital tools, often expecting instant, concise, and contextually relevant information. If AI chatbots can meet these expectations by delivering structured and comprehensible responses, users are more likely to perceive the interaction as intelligent and supportive, thereby strengthening their overall attitude toward the technology.

Additionally, the study found that the quality of information plays a crucial role in enhancing user satisfaction with AI chatbots in customer-service contexts, thereby validating Hypothesis H4. This suggests that when AI chatbots effectively address users' informational needs with reliable, up-to-date, and contextually relevant content, users report greater satisfaction with the service. This implies that high information quality not only shapes cognitive evaluation (attitude) but also enhances emotional responses (satisfaction), reinforcing the overall service experience. These results are consistent with Niu and Mvondo (2023), who emphasized that the information quality provided by AI chatbots, such as ChatGPT, plays a crucial role in enhancing user satisfaction. Similarly, Agarwal et al. (2023) discovered that information quality positively impacted satisfaction with banking chatbots in India. In the tourism field, Orden-Mejía et al. (2024) confirmed that high information quality from AI chatbots enabled users to make more informed travel decisions, thereby enhancing satisfaction. The consistency of these findings with the current study can be attributed to a universal expectation among users for informative, efficient, and credible communication in digital settings. For Generation Z in Vietnam, satisfaction arises when interactions are frictionless and intuitive, minimizing the need for additional verification or clarification. In this context, information quality becomes a core determinant of how users evaluate both the value and reliability of chatbots' interactions.

5.3. Impacts of service quality on attitude and satisfaction

The analysis results support Hypothesis H5, indicating that service quality exerts a significant and positive influence on users' attitudes toward AI chatbots within the context of service. This indicates that users tend to form positive perceptions of AI chatbots when they experience high-quality service, characterized by accurate and relevant responses, professional domain knowledge, system reliability, personalized engagement, and a modern, user-friendly interface. These elements enhance users' experience and contribute to an overall impression of the system as competent and trustworthy. This result aligns with prior research across various service domains. For example, Chowdhury (2023) emphasized the importance of service quality and convenience in shaping attitudes toward online food delivery platforms. Similarly, Prentice et al. (2020) found that service quality significantly influenced customers' attitudes toward AI-based services in the hospitality sector. More recently, Rupali et al. (2024) confirmed a significant and positive effect of service quality on user attitudes toward AI chatbots

in India. The present study extends these insights to the Vietnamese Generation Z context. This demographic is composed of digital natives who expect fast, accurate, and engaging digital interactions. In such a setting, service quality becomes a critical factor not only for task completion but also for the formation of emotional and attitudinal responses. Chatbots that offer efficient service through visually appealing and intuitive interfaces are more likely to be perceived as valuable, thereby generating stronger positive attitudes. In this context, service quality operates as both a functional and experiential driver of user acceptance.

This study also supports Hypothesis H6, demonstrating that service quality significantly enhances user satisfaction with AI chatbots. Satisfaction increases when the chatbots not only provide correct and contextually appropriate responses, but also interact in a personalized, helpful manner through a well-designed interface. This reflects users' appreciation for services that are not only efficient but also engaging and user-centered. This result is consistent with earlier findings in the field, such as Nguyen and Malik (2021), who observed that high service quality from AI-supported hotel employees led to greater customer satisfaction. Similarly, Hsu and Lin (2022) found that conversational quality and the effectiveness of AI chatbots' service recovery efforts significantly affected customer satisfaction. More recently, Yi et al. (2024) reported that service quality remained a critical factor in shaping satisfaction even in highly technical domains such as mobile trading platforms powered by generative AI. In the Vietnamese context, these results are particularly meaningful. Generation Z users have grown up with high digital exposure and are accustomed to well-designed platforms that combine both functionality and aesthetics. Their satisfaction depends not only on whether chatbots can solve problems, but also on how the interaction feels. The chatbot's tone, personalization, interface design, and response time all contribute to the overall service experience. As a result, achieving high service quality across both technical and human-like dimensions is essential for satisfying this user group within the context of swift digital transformation.

5.4. Impacts of interaction on attitude and satisfaction

The results derived from this research offer strong empirical evidence for Hypothesis H7, confirming that interaction quality with AI chatbots significantly influences users' attitudes toward these technologies in customer service contexts. Specifically, when chatbots exhibit timely responses, domain expertise, courteous communication, and personalized attention, users are more inclined to form favorable attitudes toward their use. This suggests that human-like, socially intelligent interaction contributes meaningfully to users' affective and cognitive evaluations of chatbot experiences. This finding aligns with prior studies that highlight the relational and emotional dimensions of chatbot interactions. For instance, Cao and Wen (2025) demonstrated that frequent and rich interactions with AI systems can enhance affective attitudes through perceived relational capacity. Likewise, Chakraborty et al. (2024) highlighted that perceived interaction quality positively shapes user attitudes in the setting of generative AI chatbots used for online grocery shopping. Torres et al. (2024) also found that interactivity influences both affective and cognitive attitudes toward chatbot adoption in e-commerce platforms. The alignment between these findings and this study underscores a broader understanding that the perceived human-likeness and responsiveness of AI chatbots serve as key determinants in shaping user attitudes. In the Vietnamese context, these results are particularly relevant for Generation Z, a group of digital natives that values not only speed and convenience but also engagement and social presence in digital interactions. For this cohort, chatbots that simulate natural conversation and demonstrate socially aware communication are more likely to be perceived positively, reinforcing the importance of well-designed interaction features in promoting favorable attitudes.

In contrast, this study did not find a significant relationship between

AI chatbots' interaction and user satisfaction, leading to the rejection of Hypothesis H8. This suggests that while high-quality interaction may improve users' attitudes, it is not a sufficient condition for satisfaction. Satisfaction may depend more heavily on functional outcomes such as problem resolution, efficiency, and the overall quality of service delivery rather than on relational elements alone. This finding diverges from several previous studies. For example, [Orden-Mejía et al. \(2023\)](#) identified a beneficial impact of interactivity on users' satisfaction within the tourism sector in China, while [Al-Oraini \(2025\)](#) emphasized that chatbots using socially oriented communication and demonstrating warmth and competence can enhance customer satisfaction. The difference in outcomes may stem from contextual factors, including national culture, industry expectations, and user priorities. The discrepancy may be attributed to contextual factors such as culture, generational traits, and economic development. In the case of Vietnamese Generation Z, users may value interactive features such as politeness or personalization primarily at a surface level. However, unless these are paired with reliable problem-solving and task efficiency, they are unlikely to significantly elevate satisfaction. This reflects a more pragmatic orientation, where functional performance takes precedence over emotional engagement. Furthermore, cultural expectations around emotional expression and interpersonal warmth may not be easily met by current chatbot technologies. In Vietnamese culture, meaningful interaction is often associated with genuine emotional depth, a quality that AI chatbots still lack. As such, while interactive qualities can enhance perceptions, they fall short of generating satisfaction if they do not fulfill core service expectations.

5.5. Impacts of customization on attitude and satisfaction

This study reveals that customization has a substantial effect on users' attitudes regarding AI chatbots within customer service, thus supporting Hypothesis H9. This finding suggests that when AI chatbots deliver personalized responses and adapt to users' individual preferences, they enhance users' perceived relevance, engagement, and sense of being understood, leading to more favorable attitudes. Customization appears to facilitate a user-centric experience, which is increasingly expected in modern digital interactions. This finding is consistent with existing literature. [Xia and Shannon \(2024\)](#) demonstrated a notable and positive association between customization and user attitudes in the context of online shopping, highlighting how tailored responses improve user receptiveness to chatbot services. A study by [Chi \(2024\)](#) also found that customization features in AI chatbot systems positively influenced users' pro-environmental attitudes, indicating that the benefits of personalization extend beyond commercial applications. The consistency of this effect across diverse settings suggests that customization serves as a fundamental mechanism for fostering favorable attitudes by aligning system outputs with individual user expectations. In the Vietnamese market, particularly among Generation Z, who highly value personalized digital experiences, implementing adaptive and context-aware customization features is likely to be a key driver of user acceptance and long-term usage.

Nevertheless, this study's findings show that customization does not significantly impact user satisfaction with AI chatbots, thereby leading to the rejection of Hypothesis H10. This implies that while personalized responses may shape user attitudes toward AI chatbots, they are not a determining factor in overall satisfaction. This finding suggests that, in customer service scenarios, users may prioritize task-related outcomes, such as the chatbot's efficiency to solve problems quickly and accurately to solve problems quickly and accurately, over personalization when evaluating their experience. The result corresponds with earlier studies by [Candiwan and Annikmah \(2024\)](#) and [Alnaser et al. \(2023\)](#), both of which found that customization did not have a statistically significant impact on user satisfaction within the setting of AI-driven digital banking services. These findings indicate that personalization, while valuable, may not suffice to deliver satisfaction in performance-critical

services. Conversely, [Kim and Kim \(2025\)](#) discovered that an optimal balance between standardization and customization enhances satisfaction in AI-based recommendation systems, indicating that the effect of customization is context-dependent and may be shaped by user expectations. The discrepancy in findings may be attributed to contextual and cultural differences. In Vietnam, a developing country where digital infrastructure and service standards are still maturing, user expectations may lean more heavily toward functionality and problem-solving. Among Vietnamese Generation Z users, satisfaction with AI chatbots' services is likely rooted in efficiency, accuracy, and utility rather than the degree of personalization. This cohort, although highly accustomed to digital technologies, often uses chatbots for practical and time-sensitive purposes, particularly in customer service settings. Furthermore, Vietnamese cultural norms, which may place less emphasis on individualization compared to more individualistic societies, could contribute to a lower baseline expectation for personalized digital interactions.

5.6. Impacts of problem-solving on attitude and satisfaction

The findings of this study suggest that the problem-solving capabilities of AI chatbots do not exert a statistically significant influence on users' attitudes, resulting in the rejection of Hypothesis H11. This outcome suggests that users' attitudes are not strongly shaped by the chatbot's ability to perform functional tasks such as managing returns, addressing complaints, resolving technical issues, or executing transactions. While valuable, these capabilities may be viewed as basic service expectations or tasks still better suited for human agents, particularly in more complex or emotionally charged situations. In contrast to this finding, [Chi \(2024\)](#) found that problem-solving had the strongest influence on pro-environmental attitudes when AI chatbots were used for sustainability engagement. Similarly, [Xia and Shannon \(2024\)](#) found a strong positive association between problem-solving features and user attitudes in AI chatbots' applications used for online service support. The discrepancy between those findings and this study may be explained by the contextual positioning of AI chatbots. When chatbots are used primarily as customer service tools, problem-solving may be perceived as a minimum requirement rather than a distinguishing feature that positively affects users' attitudes. Moreover, the unique characteristics of the Vietnamese Generation Z may further explain this outcome. As digital natives, this cohort often values instant results, emotional connection, and personalization over mere functionality. While they expect AI systems to solve problems efficiently, their overall attitude appears to be shaped more by how the chatbots interact, such as demonstrating empathy, social awareness, and human-like communication, than by their technical capabilities alone. Additionally, Vietnamese cultural values that emphasize emotional sensitivity and interpersonal harmony may contribute to users favoring interactive and relational experiences over task-focused functionalities. Consequently, problem-solving in isolation may not significantly influence attitudes unless it is paired with emotionally intelligent behaviors.

Conversely, this study discovers that problem-solving capabilities significantly enhance user satisfaction, thus supporting Hypothesis H12. This implies that when AI chatbots effectively understand user needs, provide appropriate solutions, and resolve issues efficiently, users perceive the chatbots as competent and reliable, leading to enhanced satisfaction with the overall service experience. This result is consistent with prior research. [Kondybayeva et al. \(2024\)](#) found that problem-solving significantly impacted satisfaction with AI voice chatbots in digital banking, while [Hsiao and Chen \(2021\)](#) identified problem-solving as a key driver of both user trust and satisfaction in food-ordering chatbots. However, contrasting evidence from [Alnaser et al. \(2023\)](#), who found no significant impact of problem-solving on satisfaction in AI-based banking services in Pakistan, suggests that contextual and cultural factors play a critical role in moderating this relationship. This discrepancy may be explained by contextual

differences. In the Vietnamese context, especially among Generation Z users, the emphasis on speed, convenience, and effective task resolution is particularly strong. This demographic tends to have high expectations for technological responsiveness and service reliability. As a result, they are likely to reward chatbots that provide swift and accurate solutions with higher satisfaction. Additionally, cultural traits such as a relatively high-power distance and a strong orientation toward efficiency and utility may further elevate the importance of competent service delivery in shaping satisfaction outcomes.

5.7. Impacts of perceived enjoyment on attitude and satisfaction

The study's results confirm a substantial positive effect of perceived enjoyment on users' attitudes toward AI chatbots within customer service settings, thereby supporting Hypothesis H13. This suggests that when users find interactions with AI chatbots to be fun, engaging, and emotionally rewarding, they are more likely to form favorable attitudes. Enjoyable experiences not only enhance engagement but also increase the perceived value of the service, emphasizing the importance of emotional resonance in addition to functional performance. These findings align with previous studies. For example, De Ciccio et al. (2020) found that perceived enjoyment significantly influenced positive attitudes toward chatbots among Italian millennials. Likewise, Kasilingam (2020) reported that perceived enjoyment significantly boosts users' attitudes in the context of shopping chatbots. Furthermore, Kim et al. (2024) demonstrated that perceived enjoyment positively affects both attitudes and usage intentions toward AI-based coding assistants. The consistency across these findings underscores the rising importance of emotional experience in digital interactions, particularly among Vietnamese Generation Z users, who are digital natives. This generation tends to prioritize not only efficiency but also engaging and emotionally enriching interactions. Their preference for emotionally responsive technology, shaped by both generational traits and cultural emphasis on user experience, likely reinforces the strong connection between enjoyment and positive attitudes toward AI chatbots.

Besides, this research reveals that perceived enjoyment significantly enhances user satisfaction with AI chatbots, further supporting Hypothesis H14. When users experience enjoyable and emotionally satisfying interactions, they are more inclined to express a higher level of overall satisfaction with the service, even in the absence of human contact. Positive emotional responses contribute to a more meaningful and memorable user experience, strengthening the relationship between the user and the AI service. This insight aligns with prior scholarly research. Specifically, Huang et al. (2024) demonstrated that perceived enjoyment significantly predicts both satisfaction and future usage intentions of AI systems in hospitality and tourism. Likewise, Orden-Mejía et al. (2024) reported a positive association between perceived enjoyment and satisfaction with AI-based destination chatbots in Ecuador. Furthermore, Pereira et al. (2021) confirmed the critical role of enjoyment in fostering customer satisfaction across various service contexts. The alignment of these results reinforces the notion that emotional gratification is a key driver of user satisfaction in digital environments, particularly among younger generations. In the Vietnamese context, Generation Z users expect AI chatbots not only to perform tasks efficiently but also to provide a pleasant and emotionally resonant experience. This generational expectation, combined with cultural tendencies that favor emotional connection and user-centered design, likely strengthens the observed relationship between enjoyment and satisfaction in AI chatbots' interactions.

5.8. Impacts of attitude on continuance intention

This research outcome provides empirical evidence for the substantial impact of user attitude on users' intention to continue using AI chatbots in customer service, thereby confirming Hypothesis H15. When users perceive AI chatbots as useful, enjoyable, and engaging, such as

making service interactions more interactive or entertaining, they are more likely to maintain their use over time. This outcome is consistent with preceding literature, as Xu et al. (2024) confirmed that a positive attitude is a strong predictor of continuance intention among users of AI painting tools in China. Similarly, Huang and Lee (2022) emphasized the pivotal role of user attitude in promoting sustained use of fintech chatbots. In a different context, Pasupuleti and Thiyyagura (2024) demonstrated that attitude was a key predictor of long-term usage intentions among higher education students utilizing ChatGPT. The consistency of these results across studies underscores the central role of attitude in shaping user behavior in digital service environments. Within the context of Vietnam's rapidly growing digital economy, Generation Z users, who are typically tech-savvy and highly engaged with digital innovations, tend to value services that are both efficient and emotionally engaging. When AI chatbots are perceived not only as functional tools but also as enjoyable and interactive companions, users tend to form more positive attitudes, which subsequently fosters stronger intentions for continued service usage.

5.9. Impacts of satisfaction on continuance intention

This study substantiates the significant role of user satisfaction in influencing continuance intention toward AI chatbots in customer service contexts, thereby supporting Hypothesis H16. Specifically, when users perceive that AI chatbots meet their service quality expectations, improve their brand experience, and create a sense of being acknowledged and appreciated, they tend to maintain long-term usage. This finding is in line with several prior research. For instance, Esmailzadeh et al. (2025) found that user satisfaction significantly influences continuance intention in the context of seeking health-related information through AI chatbots in the United States. Similarly, Alvionita (2024) identified that satisfaction has a substantial impact on customers' intention to continue using chatbots in e-marketplaces. These findings highlight the pivotal role of satisfaction in the adoption and continued use of digital services. In Vietnam, Generation Z users are characterized by their digital fluency and high expectations for seamless, efficient service experiences. When AI chatbots deliver prompt, accurate, and contextually appropriate responses, they meet these expectations, thereby increasing user satisfaction. In turn, this satisfaction serves as a strong motivator for ongoing engagement with AI chatbots in the evolving digital landscape of Vietnam.

6. Limitations and recommendations

While offering valuable insights, this study acknowledges several limitations that should be addressed in future research. Firstly, this study employed a non-probability convenience sampling method with a relatively limited sample size, which may place constraints on the generalizability of the findings. The sample might not adequately reflect the wider population of AI chatbot users. Future research should target a broader and more varied sample population that includes participants across different age groups, geographic regions, education levels, and socio-economic backgrounds, to strengthen the external validity of the findings and facilitate more comprehensive comparisons across user segments. Secondly, this study utilized self-reported data, which may be subject to potential biases such as social desirability, recall inaccuracies, or subjective misjudgments. These biases can potentially affect the reliability and objectivity of the results. To mitigate this limitation, future studies should consider integrating multi-method approaches, such as behavioral analytics, experimental methods, and qualitative interviews, to triangulate data and gain a more nuanced understanding of user interactions with AI chatbots. Thirdly, this study did not examine in depth the potential impact of demographic factors such as digital literacy or educational background, which could significantly influence users' perceptions and behavioral responses toward AI chatbot use. Future studies should investigate how such characteristics interact with

key variables like usability and trust to better understand diverse user needs and adoption patterns. Finally, this study concentrated on the use of AI chatbots within a general customer service context. It did not account for industry-specific applications, such as in healthcare, finance, or education, where user expectations and experiences may differ substantially. Future investigations should examine sector-specific contexts to generate more targeted insights and inform the design of AI chatbots tailored to the unique requirements of each industry. Addressing these limitations will contribute to the development of more comprehensive, generalizable, and context-sensitive frameworks, ultimately supporting the effective implementation of user-centered AI chatbot systems across various digital service environments.

7. Implications

7.1. Theoretical implications

This study offers several valuable theoretical contributions, bridging the fields of information systems, consumer behavior, and digital marketing. By extending the ISSM, originally developed to assess system-level effectiveness, with chatbot marketing efforts constructs such as interaction, customization, problem-solving, and perceived enjoyment, the research advances a more integrated understanding of user experience in AI-mediated service environments. This multidimensional approach enriches the literature by demonstrating that, beyond technical efficiency, users' affective experiences and engagement behaviors play critical roles in shaping post-adoption outcomes such as satisfaction, attitude, and continuance intention using AI chatbots.

Firstly, this study strengthens the theoretical foundation of the ISSM by confirming the central role of its three core constructs: system quality, information quality, and service quality, in influencing both user attitude and satisfaction. Unlike earlier studies that often focused on system use or user satisfaction in general IS contexts, these findings demonstrate that in the post-adoption stage of AI chatbots' usage, these quality-related attributes remain critical. Their dual impact suggests that users simultaneously evaluate the reliability, responsiveness, and informativeness of chatbot systems from both functional and emotional standpoints. This reinforces the validity and versatility of the ISSM when extended into newer, user-facing technologies such as conversational AI.

Secondly, beyond the ISSM framework, this study also reveals important insights into how chatbot-specific marketing effort variables shape user experience through distinct psychological mechanisms. Interaction and customization influence only user attitude, indicating their affective nature and their role in generating perceived personal relevance and social presence. In contrast, problem-solving significantly enhances only satisfaction, suggesting that users appraise it as a pragmatic, task-oriented function. Notably, perceived enjoyment exerts a dual effect on both attitude and satisfaction, positioning it as a bridging factor between hedonic experience and service evaluation. These distinctions support the understanding of user responses to AI-driven services and provide a conceptual basis for further theoretical exploration of emotional versus functional value in technology adoption and continuance.

Thirdly, by focusing on Generation Z, whose digital fluency and technology-related expectations differ significantly from previous generations, this study advances theoretical understanding of age-based segmentation in technology adoption. Their preference for immediacy, personalization, and seamless interaction highlights the importance of emotionally engaging and responsive experiences. The finding that this cohort responds more strongly to customization, interaction, and perceived enjoyment suggests that continuance intention is shaped by both functional and affective factors. Additionally, by situating the research in Vietnam, this study yields significant insights into how cultural context influences consumer attitudes toward AI technologies, emphasizing the need to incorporate cultural dimensions into future technology adoption models.

Lastly, this study makes contributions to the literature by developing and empirically validating an integrated framework that links system-related attributes and user experience factors through the mediating roles of satisfaction and attitude to explain users' continuance intention. By capturing both functional and affective dimensions of user evaluation, the model reflects the complexity of post-adoption behavior in AI-driven service environments. This theoretical advancement provides a solid foundation for future research to investigate long-term behavioral outcomes such as sustained engagement, loyalty, and word-of-mouth. It is particularly relevant in digitally saturated, experience-centric markets where technological effectiveness must be complemented by emotional resonance to maintain user retention.

7.2. Practical implications

The findings of this study generate several important practical implications for stakeholders involved in the development, deployment, and strategic use of AI chatbots, particularly in consumer-facing industries targeting Generation Z.

Firstly, for developers and user experience (UX) designers, the significant influence of system quality, information quality, and service quality on both user attitude and satisfaction emphasizes the critical need for technical reliability, functional accuracy, and responsive service delivery. Ensuring smooth conversational flow, minimal response latency, and the provision of relevant, coherent information are foundational to fostering continued usage. Moreover, the impact of perceived enjoyment and customization on user attitude suggests that AI chatbots' design should move beyond technical efficiency to include emotionally engaging features. Developers should integrate adaptive responses, natural language variation, and design elements that allow users to feel recognized and valued throughout the interaction process.

Secondly, from a marketing and customer experience perspective, the results indicate that experiential attributes such as customization, interaction, and perceived enjoyment play a substantial role in shaping users' affective responses. These findings suggest that AI chatbots should be strategically positioned not only as service tools but also as interactive brand interfaces that reflect personality, tone, and values aligned with the target audience. Marketing teams can enhance engagement by emphasizing personalized dialogue, contextual awareness, and emotionally resonant messaging in AI chatbots' communications. Such approaches may be particularly effective for Generation Z users, who value authenticity, interactivity, and enjoyment in digital service encounters.

Thirdly, for business managers and strategic decision-makers, the dual mediating roles of satisfaction and attitude highlight the importance of delivering holistic AI chatbot' experiences that address both utilitarian and hedonic user needs. While features such as problem-solving primarily enhance satisfaction by resolving practical concerns, emotional and relational features drive more favorable attitudes that underpin long-term engagement. Therefore, organizations should allocate resources not only to backend infrastructure and automation but also to front-end experience design and continuous improvement based on user feedback. Furthermore, AI chatbots' implementation should be regarded as a strategic long-term investment in managing customer relationships, rather than a short-term cost-reduction strategy, particularly in digitally saturated markets where service quality and user experience can become sources of competitive differentiation.

Finally, the results of the IPMA provide clear, data-driven priorities for managerial decision-making. While system quality, information quality, service quality, and perceived enjoyment are highly important for continuance intention, system quality shows the strongest overall impact. However, their current performance levels reveal room for improvement. Organizations should consider reallocating resources to enhance these areas. This may include improving system reliability, optimizing response speed, refining chatbots' content, and incorporating more engaging and emotionally resonant design features. Focusing on

these key factors can significantly strengthen user satisfaction and long-term engagement with AI chatbots.

8. Conclusion

This study investigated the determinants influencing continuance intention toward AI chatbot services among Vietnamese Generation Z by extending the ISSM with chatbot-specific marketing effort constructs. Using data from 562 valid responses and analyzed through PLS-SEM, this study offered a nuanced understanding of both functional and experiential determinants of post-adoption behavior.

The findings revealed that system quality, information quality, service quality, and perceived enjoyment significantly influenced both attitude and satisfaction, underscoring that Generation Z users evaluated AI chatbots not only on their technical capabilities but also on emotional engagement. Among these, system quality emerged as the most influential factor, emphasizing its foundational role in shaping user perceptions. Customization and interaction positively affected attitude, enhancing perceived relevance and user engagement, while problem-solving contributed solely to satisfaction, highlighting its importance in addressing users' functional needs. Importantly, both attitude and satisfaction strongly predicted continuance intention, reinforcing the need for a holistic strategy that incorporated both performance and emotional experience to foster sustained use of AI chatbots.

From a theoretical perspective, this study extended the ISSM framework through the integration of affective and experiential constructs, validating a dual-pathway model where both cognitive (attitude) and emotional (satisfaction) responses mediated the correlation between system attributes and continued usage. More importantly, by emphasizing Generation Z in Vietnam, this study provided contextual insights into how cultural and generational factors shape technology adoption, especially in a digitally native and developing country. From a practical perspective, the results highlighted the importance of optimizing system reliability, informative content, and service responsiveness while embedding emotionally engaging features such as enjoyment, interaction, and personalization. Marketers and developers were advised to focus on creating AI chatbot experiences that are not only efficient but also enjoyable and personally relevant to retain the attention and loyalty of digitally fluent Generation Z users.

CRedit authorship contribution statement

Thi Thuy An Ngo: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Thi Yen Nhi Phan:** Writing – review & editing, Writing – original draft, Visualization, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Trung Khoi Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Nguyen Bao Tran Le:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ngoc Thien An Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Thi Thuy Duong Le:** Writing – review & editing, Writing – original draft, Visualization, Resources, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

Data supporting the results of this study can be obtained from the corresponding author on request.

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